

AASC STANDARD MODEL SYMMETRY ARC

SYM-5

Flavor Symmetry, Basis Skin, Spurion Structure, and Invariant Normal Form

A Target-Fibered Orbit–Stabilizer–Invariant and Role-Occupancy Closure

Release status. Audit-grade integrated manuscript and executable certificate. The framework-neutral flavor action, basis quotient, fixed-tensor stabilizers, generic and registered singular strata, invariant and generalized-CP packages, conditional MFV fibers, higher-flavor interface, AASC role classification, rival ledger, and finite role-occupation certificate are synchronized in one reproducible project.

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Document control, abstract, and results at a glance

Control record	
Arc module	SYM-5, importing the frozen SYM-0 through SYM-4 theorem interfaces.
Project release	v1.0.0, audit-grade integrated manuscript and executable certificate.
Source package	SYM 5 Source.zip, the post-matrix Standard Model Carrier ARC, the completed SYM manuscript bundle, and a primary QFT source dossier.
Authority rule	Substantive lineage and explicit supersession govern over copy time. The full and manuscript-only SYM-3 archives are packaging profiles of one frozen theorem authority.
Endpoint	Kernel-Faithful Integrated Flavor Action–Quotient–Spurion–Invariant Occupancy Closure.
Local closure	Eight sharp-local or EFT target fibers, twelve registered generic/singular strata, and one exact higher-flavor interface fiber.
Executable audit	82 independently constructed role burdens, 82 independently constructed QFT residues, 778 evaluated same-target pairs, one compatible assignment, one lawful assignment orbit, and twelve mutation tests.
Global boundary	Full line/defect/global-form closure remains SYM-9; whole-arc symmetry exhaustion remains SYM-11.

Abstract

This paper gives a target-indexed classification of Standard Model flavor structure that keeps six objects rigorously distinct: the zero-Yukawa kinetic flavor action, weak-basis redescription, the stabilizer of a fixed Yukawa or mass-tensor point, spurion covariance, quotient-stable mass/mixing/CP invariants, and anomaly or higher-group data. The central mathematical object is the action groupoid

$$\mathcal{Y}_{\mathcal{D}}//G_{\mathcal{D}}^{\text{basis}},$$

formed separately in the minimal zero-Yukawa Standard Model, the generic and singular quark sectors, the minimal massless-neutrino lepton sector, Dirac-neutrino and Weinberg/Majorana neighboring targets, quark and minimal-lepton MFV EFTs, and a higher-flavor interface target. The physical flavor symmetry at a fixed tensor point is proved to be its stabilizer, not the complete covariance group. Generic parameter counts follow from tensor-space dimension minus basis-group

dimension plus stabilizer dimension; registered singular strata are recomputed rather than inherited from a generic CKM or PMNS chart. Quark, Dirac-lepton, and Majorana generalized-CP conditions are stated in target-specific form, and invariant completeness is indexed by target and orbit stratum. Spurions organize breaking ancestry and EFT operator grammar without becoming dynamical or primitive objects. The AASC classification is applied only after the framework-neutral QFT construction is complete. It imports the independent carrier result that flavor-transition standing and CP/holonomy residue have primitive support, while bases, matrix entries, chart phases, and labels do not thereby acquire primitive authority. The final finite certificate independently constructs 82 role positions and 82 QFT residues, evaluates all 778 same-target pairs, preserves the relation graphs, and leaves one complete lawful assignment orbit. The result closes the declared sharp-local SYM-5 flavor architecture without predicting numerical Yukawa data, deriving generation cardinality from matrix size, asserting MFV as a fact of nature, or pre-empting SYM-9 global-form and defect closure.

Central decomposition

$$\begin{aligned} \text{flavor architecture} = & \text{limit action} + \text{basis groupoid} + \text{fixed-point stabilizer} \\ & + \text{breaking tensors} + \text{physical invariants} + \text{quantum/frontier status.} \end{aligned}$$

The fixed-point stabilizer is the term that prevents basis covariance from being misreported as a physical symmetry of a fixed numerical tensor point.

Results at a glance

Result	Audit-grade conclusion
Zero-Yukawa group	The maximal local unitary commutant is the $U(3)^5$ presentation cover. The faithful global action quotients the gauged hypercharge direction and a global-form-dependent central kernel; Abelian and anomaly statuses are separately recorded.
Basis versus symmetry	G_D^{basis} acts on tensor space and generates representative arrows. The actual symmetry of a fixed point y is $\text{Stab}_{G_D^{\text{basis}}}(y)$.
Generic quark/Dirac count	For n generations, $\dim \mathcal{Q}_{\text{phys}} = n^2 + 1 = 2n + n(n-1)/2 + (n-1)(n-2)/2$. At $n = 3$: six masses, three angles, one phase.
Generic Majorana count	For the low-energy Weinberg target, $\dim \mathcal{Q}_{\text{phys}} = n(n+1)$. At $n = 3$: six masses, three angles, three CP phases.
Singular strata	Twelve explicit strata carry tensor dimension, full-pair stabilizer dimension, quotient dimension, factorization isotropy, CP package, and successor status.
Invariant normal form	A finite quark invariant package and target-specific lepton packages separate physical data at their declared scope; generic and singular completeness are not conflated.
MFV	Quark and minimal-lepton MFV close conditionally under an explicit complete-spurion hypothesis and EFT truncation. Spurions remain nondynamical backgrounds.
Higher flavor	The zero-form/one-form higher-group structure is transported as an exact conditional interface; its global-form, monopole, line, and defect completion remains SYM-9.

Result	Audit-grade conclusion
AASC role closure	Limit actions, fixed-point actions, basis skin, tensors, spurion routes, invariant witnesses, obstructions, interfaces, and target updates receive distinct locus-sensitive roles.
Finite certificate	9 fibers, 25 results, 12 strata, 82 roles, 82 residues, 778 evaluated pairs, 1 complete assignment, 1 lawful orbit, 12/12 mutation tests passed.

Reading rule

Every substantive claim follows the same order: fixed target; framework-neutral QFT object; action and quotient; stabilizer and invariant certificate; anomaly, scale, and global-form fields; then AASC role and rival disposition. The mainstream ledger remains assessable independently of the AASC interpretation, while the AASC dependency order remains antecedent to any determinate use of the QFT objects.

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Strongest realistic endpoint and proof strategy

1.1 Endpoint package

The ARC assigns SYM-5 the task of separating the $U(3)^5$ limit symmetry, basis redundancy, spurions, mixing representatives, physical invariants, degeneracy stabilizers, neutrino variants, and a higher-flavor interface [1]. The completed SYM-0–SYM-4 sequence and the later Carrier ARC permit a stronger endpoint: the complete target-fibered flavor action–quotient–stabilizer–spurion–invariant architecture, followed by a locus-sensitive role normal form and a finite assignment-orbit certificate.

Kernel-Faithful Integrated Flavor Action–Quotient–Spurion–Invariant Occupancy Closure

Fix the registered target family $\mathfrak{D}_5$ and the sharp-local operator categories declared in the target registry. For every closed fiber $\mathcal{D} \in \mathfrak{D}_5^{\text{closed}}$ and every registered orbit-type stratum λ :

1. the kinetic-limit action, basis groupoid, tensor space, fixed-point stabilizer, physical quotient, invariant package, generalized-CP condition, anomaly status, and stratum are fully typed;
2. representative transformations descend to one physical quotient point exactly when they are lawful weak-basis arrows preserving the complete target;
3. every flavor-relevant object occupies one or more compatible loci in the finite role vocabulary;
4. independently constructed role positions and independently constructed quotient-stable QFT residues admit exactly one complete lawful assignment orbit;
5. every same-scope rival is absorbed as equivalent, rejected as skin-only, redirected as a target update or interface, or identified as a theorem-breaking counterexample.

The higher-flavor defect-completed fiber is excluded from the sharp-local quantifier until SYM-9 fixes the faithful global form, line/defect category, monopole sectors, background fields, and higher-group extension data.

1.2 Formal endpoint statement

For each closed target/stratum pair $(\mathcal{D}, \lambda)$ let

$$\text{Role}_{\mathcal{D}, \lambda}, \quad \text{Res}_{\mathcal{D}, \lambda}$$

be the independently authored role positions and QFT residues. Let $\text{Elig}_{\mathcal{D},\lambda}$ be the complete evaluated pair relation, $\text{Comp}_{\mathcal{D},\lambda}$ the relation-preserving compatibility predicate, and $\Gamma_{\mathcal{D},\lambda}^{\text{lawful}}$ the automorphism group preserving target, kind, semantic signature, source, and relation fields. The executable endpoint is

$$\left| \frac{\text{Assign}_{\mathcal{D},\lambda}(\text{Role}_{\mathcal{D},\lambda}, \text{Res}_{\mathcal{D},\lambda}; \text{Elig}_{\mathcal{D},\lambda}, \text{Comp}_{\mathcal{D},\lambda})}{\Gamma_{\mathcal{D},\lambda}^{\text{lawful}}} \right| = 1.$$

The product over all target blocks also has cardinality one. This is a statement about classification closure, not a preference for one weak basis, one CKM/PMNS chart, or one numerical tensor point.

1.3 Proof architecture

The proof has six layers.

1. **Target freeze.** Nine fibers and twelve orbit strata fix field content, operator set, tensor law, basis group, global-form boundary, scale, and evidence identity.
2. **Framework-neutral construction.** The commutant, groupoid, stabilizers, quotient dimensions, invariants, CP maps, anomalies, and spurion grammars are constructed without AASC role premises.
3. **Target transport.** Basis arrows, parameter-limit transports, EFT updates, and field/global-form updates are separated.
4. **AASC typing.** Each QFT object is assigned a locus-sensitive role only after its quotient and standing-bearing content are fixed.
5. **Finite closure.** Separate constructor files generate roles and residues; all 778 same-target pairs are evaluated; relation graphs and source fields are checked; the one surviving bijection is quotiented by lawful automorphisms.
6. **Adversarial closure.** Ten rival classes and twelve mutation tests attack target, anomaly, invariant, interface, evidence, independence, and label-invariance fields.

1.4 Strength and boundary

The endpoint is stronger than a descriptive flavor survey because it proves the exact location of the symmetry/basis distinction, recomputes singular strata, supplies target-relative invariant completeness, and integrates an executable role certificate. It is deliberately narrower than an arbitrary flavor-model classification. It does not select numerical Yukawa values, derive three generations from matrix size, make MFV unconditional, close arbitrary seesaw invariant rings, or claim the defect-completed Standard Model before SYM-9.

Endpoint status

The theorem registry contains 25 completed, imported-and-specialized, conditional, interface, or finite-certificate results. No entry retains a planned or skeleton status. The final theorem, SYM5-T25, is proved at the declared sharp-local scope; its higher-flavor and whole-arc exclusions are part of the theorem statement rather than post hoc qualifications.

Source-package audit and authority resolution

2.1 Archive audit

The supplied `SYM 5 Source.zip` passed ZIP integrity testing. Its 37 top-level artifacts comprise 21 complete source/project archives and 16 PDFs. The nested projects contain 2,706 regular files: 600 TeX sources, 207 CSV ledgers, 788 JSON artifacts, 108 PDFs, 23 Lean files, 273 Markdown files, and 7 BibTeX databases. The exact file-level audit is preserved in `data/source_package_audit.csv` and JSON form.

The package is strongest on the completed SYM status/anomaly interfaces, the AASC dependency stack, CNF/MGN Yukawa–texture–spectrum–mixing machinery, neutrino reconstruction, and the Standard Model representation/Yukawa architecture. Primary QFT papers supplement it for the $U(3)^5$ presentation, finite invariant algebra, MFV, Majorana CP conditions, and higher flavor. The primary-source map records the technical use and scope guard of each external result.

2.2 Post-matrix Carrier ARC authority

The corpus matrix predates the completed Standard Model Carrier ARC. The later Carrier ARC therefore governs primitive-support and no-promotion claims. Its flavor endpoint supplies primitive support for the flavor-transition and CP/holonomy-residue roles while refusing primitive authority to raw CKM/PMNS entries, weak-basis charts, phase coordinates, diagonalizations, and flavor labels. The separate generation-cardinality endpoint supplies the fixed three-generation governance envelope. Consequently,

3×3 tensor notation $\nRightarrow$ a proof of three generations,

and

primitive flavor-transition support $\nRightarrow$ a preferred basis or numerical mixing matrix.

SYM-5 uses the Carrier ARC as support beneath the QFT quotient; it does not use carrier labels to manufacture the quotient.

2.3 SYM-3 packaging resolution

The previously observed SYM-3 hash difference is resolved by the user’s authority statement: the full project archive and the manuscript-only Overleaf archive are packaging profiles of the same

frozen theorem object. The manuscript-only package was created because the full archive exceeded Overleaf’s upload limit. No theorem, definition, evidence identity, or downstream interface changed. The release therefore records one mathematical authority with multiple packaging hashes rather than a successor chain.

SYM-3 disposition

Disposition: packaging-equivalent, theorem-identical. The hash difference does not trigger a mathematical rereview, target update, or SYM-4 regression. The import manifest records SUPPLIED_FULL_AND_MANUSCRIPT_ONLY_PACKAGING.

2.4 SYM-4 theorem interface

The supplied SYM-4 article and Technical Audit Companion freeze the immediate handoff: the weak-basis and rephasing quotient, generalized-CP existence conditions, target-relative CP-invariant packages, degeneracy stabilizers, neutrino target bifurcation, and conservative successor transport [2]. SYM-5 imports these results and expands them into the full flavor action groupoid. It neither reopens the discrete ledger nor silently strengthens a target-relative invariant statement into a universal one.

2.5 Primary-source completion

The load-bearing primary sources are mapped in Appendix C. Their uses are narrowly assigned:

- D’Ambrosio–Giudice–Isidori–Strumia for the $U(3)^5$ flavor presentation, quark spurions, and the MFV hypothesis [3];
- Jarlskog and Jenkins–Manohar for quark CP and rephasing/invariant structure [4–6];
- Hanany–Jenkins–Manohar–Torri for Hilbert-series control of finite invariant algebras [7];
- Cirigliano–Grinstein–Isidori–Wise for the selected minimal-field-content lepton MFV target [8];
- del Aguila–Zralek and Yu–Zhou for vanishing/degenerate masses and Majorana CP conditions [9–11];
- Córdova–Koren for the higher-flavor zero-form/one-form interface [12];
- Weinberg for the dimension-five Majorana target [13].

No review article or repeated corpus assertion substitutes for a primary source when a primary theorem is available.

2.6 Audit conclusion

The source audit authorizes the full integrated release. All previously identified source-level blockers are resolved or converted into explicit theorem boundaries. The remaining global-form and defect data are not missing inputs to a falsely broad local theorem; they are the declared input surface of the SYM-9 interface.

Dependency order and imported theorem surface

3.1 Locked dependency order

The project preserves two complementary orders.

Kernel-first in necessity

A non-degenerate determinate flavor target already presupposes admissibility, standing, reference, and irreversible fixation. The AASC kernel is therefore not an optional interpretive layer applied after independently complete objecthood [14].

QFT-first in certification

The particular fields, tensors, group actions, quotient maps, stabilizers, invariants, anomalies, and target changes used by SYM-5 are nevertheless constructed and certified in framework-neutral QFT and mathematics before an AASC role is assigned. Kernel necessity does not substitute for positive flavor construction.

The combined order is

determinate target $\Rightarrow$ kernel instantiation $\Rightarrow$ independent QFT construction
 $\Rightarrow$ quotient/invariant certificate $\Rightarrow$ role classification $\Rightarrow$ finite closure.

3.2 Completed SYM imports

Module	Frozen import	Use in SYM-5
SYM-0	Typed status record, quotient-action trichotomy, role covariance, target update, independently formed positions/residues, lawful assignment orbit	Supplies the record grammar and finite-certificate architecture. It does not pre-determine flavor-specific roles.
SYM-1	Gauge/representation scaffold, local product, representation response, anomaly-free architecture	Fixes which species have equivalent gauge representations and may therefore be mixed by a flavor action.

Module	Frozen import	Use in SYM-5
SYM-2	Active/passive discipline and CPT premise bundle	Prevents passive description from being conflated with physical action and governs CP/CPT compatibility through SYM-4.
SYM-3	Gauge, global-current, global and 't Hooft anomaly distinctions; conservative transport	Types Abelian flavor currents and the higher-flavor interface without collapsing anomaly meanings.
SYM-4	Basis/rephasing quotient, generalized CP, invariant packages, degeneracy and neutrino bifurcation	Immediate discrete-symmetry handoff; SYM-5 embeds it in the full flavor orbit–stabilizer architecture.

3.3 Carrier ARC import

The Carrier ARC supplies primitive support, not a substitute classifier. It establishes that flavor-transition and CP/holonomy-residue roles are supported by the fixed carrier architecture; it does not assign a privileged matrix representative. SYM-5 therefore has two separate obligations:

1. construct the quotient-stable tensors, spectra, relative eigenspace data, and CP residues that realize the supported roles;
2. prove that bases, matrix entries, phase charts, diagonalization choices, and labels have no primitive authority once the quotient-stable package is fixed.

3.4 Foundation imports

The kernel, fixed-domain exhaustion, Anchor–Tensor–Skin decomposition, same-scope closure, and role-occupancy package provide the admissibility and finite-classifier logic [14–18]. Their imported consequences are used with exact theorem boundaries:

- a representative difference that changes no admitted standing, continuation, or quotient predicate is skin;
- a coverage-, transformation-, or continuation-bearing difference is tensor or action data, not skin;
- target updates are not same-scope counterexamples;
- role positions and candidate residues must be formed independently before compatibility filtering;
- one lawful assignment orbit is a classification endpoint, not a numerical dynamics theorem.

3.5 No authority reversal

Three invalid reversals are explicitly blocked:

$$\begin{aligned} \text{AASC role label} &\not\Rightarrow \text{QFT group action,} \\ \text{formal QFT covariance} &\not\Rightarrow \text{primitive carrier status,} \end{aligned}$$

and

machine-checked record consistency $\nRightarrow$ proof of imported QFT premises.

The manuscript can therefore be audited at three distinct layers: source theorem, project record, and AASC consequence.

Target family, fingerprints, and transport

4.1 Why a fibered family is mandatory

An undifferentiated “Standard Model flavor target” would combine incompatible field content, tensor types, rephasing freedoms, parameter counts, lepton-number statuses, and anomaly statements. SYM-5 instead fixes a fibered family $\mathfrak{D}_5$. Each fiber carries the fingerprint

$$\mathfrak{f}(\mathcal{D}) = (F_{\mathcal{D}}, \mathcal{O}_{\mathcal{D}}, \mathcal{Y}_{\mathcal{D}}, G_{\mathcal{D}}^{\text{basis}}, \mathcal{B}_{\mathcal{D}}, \mathcal{G}_{\mathcal{D}}^{\text{global}}, \lambda_{\mathcal{D}}, E_{\mathcal{D}}),$$

where F is field content, $\mathcal{O}$ the admitted operator set, $\mathcal{Y}$ the tensor space, $\mathcal{B}$ backgrounds and boundaries, $\mathcal{G}^{\text{global}}$ global-form/extended-operator data, λ the orbit stratum, and E the evidence version.

4.2 Frozen target registry

ID	Target	Closure status / boundary
D5-0Y-MIN	Zero-Yukawa minimal renormalizable Standard Model	CLOSED WITH GLOBAL FORM BOUNDARY: nonzero Yukawas or added neutrino operator create successor target
D5-Q-GEN	Generic three-generation quark Yukawa target	CLOSED: rank loss, degeneracy or alignment moves to D5-Q-STRAT
D5-Q-STRAT	Registered singular and degenerate quark strata	CLOSED FOR REGISTERED STRATA: unregistered orbit type requires a versioned successor row
D5-L-0NU	Minimal renormalizable lepton target with massless neutrinos	CLOSED: nu_R or Weinberg operator changes target
D5-L-DIRAC	Dirac-neutrino neighboring target	CLOSED: Majorana mass or Weinberg operator changes target
D5-L-MAJ5	Dimension-five Majorana/Weinberg effective target	CLOSED FOR REGISTERED STRATA: explicit seesaw completion is a new target

ID	Target	Closure status / boundary
D5-MFV-Q	Quark Minimal Flavor Violation EFT target	CONDITIONAL CLOSURE: new flavor-breaking tensor creates a non-MFV successor
D5-MFV-L	Minimal-field-content lepton MFV EFT target	CONDITIONAL CLOSURE: extended-field-content seesaw MFV is a distinct target
D5-HF-IFACE	Higher-flavor/global-form interface	INTERFACE ONLY TO SYM9: SYM-9 fixes the complete defect/global-form target

The first eight fibers are included in the local or conditional closure quantifier according to their recorded status. **D5-HF-IFACE** is a well-typed interface fiber but is excluded from sharp-local closure. This is not an unresolved local row: the required line, monopole, and global-form data change the target.

4.3 Target-fingerprint theorem

SYM5-T01 – Target-family and fingerprint completeness

Relative to the frozen candidate-generation protocol, every theorem, atlas row, invariant package, CP criterion, role, and assignment record carries a target fingerprint sufficient to determine field content, operator set, tensor law, basis group, stratum, global-form boundary, scale, and evidence identity. A change in any of these fields is either a registered parameter-stratum transport or a versioned target successor.

Proof / certificate

The target registry contains nine unique IDs. Every theorem, stratum, role, QFT residue, invariant, CP map, spurion, interface, and atlas row is validated against those IDs. The target-transport registry exhausts all inter-fiber maps used by the manuscript: zero-Yukawa activation, generic-to-singular quark transport, minimal-to-Dirac field update, minimal-to-Majorana EFT update, two MFV framework updates, and the global-form/background update to higher flavor. The validator rejects unknown targets and hidden neutrino updates. This proves completeness relative to the declared protocol; it does not claim completeness over arbitrary BSM targets.

4.4 Four transport classes

A map $\tau : \mathcal{D} \rightarrow \mathcal{D}'$ receives exactly one of the following statuses in the project:

1. **representative transport**: the full fingerprint and physical quotient point are fixed; only coordinates change;
2. **parameter-stratum transport**: field and operator content are fixed but rank, degeneracy, alignment, or another orbit-type datum changes;
3. **EFT target update**: the operator set, truncation, or matching scale changes, as for the Weinberg operator or MFV hypothesis;
4. **field/global-form update**: fields, faithful gauge group, line/defect category, monopoles, or backgrounds change.

Only the first is skin by construction. The second can enlarge factorization isotropy or the physical stabilizer. The third and fourth create successor target records.

4.5 Candidate-generation protocol

The finite inventory is generated by traversing:

- unitary commutants of the fixed gauge representation and kinetic operator;
- every lawful action on the registered Yukawa or mass tensors;
- generic and named singular orbit strata;
- invariant functions, generalized-CP maps, currents, and anomalies;
- spurion/EFT operator grammars;
- massless, Dirac, and Majorana neutrino targets;
- global-form and extended-operator interfaces;
- every AASC role, same-scope rival, and downstream transport used by the paper.

A proposed object outside this protocol either exposes an omitted same-scope locus or changes target. The rival protocol requires the objector to show which alternative applies.

Framework-neutral zero-Yukawa flavor architecture

5.1 Gauge representation and multiplicity spaces

Use the right-handed-field convention

$$Q_L, \quad u_R, \quad d_R, \quad L_L, \quad e_R,$$

with three copies of each species. The fermion representation decomposes as

$$\mathcal{H}_f = \bigoplus_{f \in \{Q, u, d, L, e\}} V_f^{\text{gauge}} \otimes \mathbb{C}_f^3,$$

where the V_f^{gauge} are pairwise inequivalent Standard Model gauge representations. The kinetic operator and gauge action are proportional to the identity on each multiplicity factor $\mathbb{C}_f^3$.

5.2 Commutant theorem

SYM5-T03 – Zero-Yukawa commutant theorem

For the minimal three-generation Standard Model with all Yukawa tensors set to zero, the maximal local unitary commutant of the gauge representation and kinetic operator is the presentation cover

$$G_F^{\text{cl}} = U(3)_Q \times U(3)_u \times U(3)_d \times U(3)_L \times U(3)_e.$$

No lawful unitary commutant mixes inequivalent gauge species.

Proof / certificate

A unitary operator commuting with the gauge action preserves each isotypic component. Because the five V_f^{gauge} are inequivalent, the commutant is block diagonal. On $V_f^{\text{gauge}} \otimes \mathbb{C}_f^3$, Schur's lemma gives identity on the irreducible gauge factor tensored with an arbitrary endomorphism of the multiplicity space. Unitarity reduces that endomorphism to $U(3)_f$. The direct product over the five blocks gives $U(3)^5$. The result is the maximal local presentation cover; faithfulness and current anomalies are separate questions. This specializes the standard MFV flavor presentation [3] to the fixed SYM-5 target.

5.3 Presentation cover versus faithful global action

The product $U(3)^5$ is not yet the final faithful global group. A continuous one-parameter subgroup reproduces the gauged hypercharge action on the fermions. In addition, discrete center elements can coincide with gauge-center elements; the precise discrete kernel depends on the faithful global Standard Model gauge group. Accordingly, the sharp-local result is

$$G_F^{\text{faith}} = \frac{U(3)^5}{U(1)_Y^{\text{gauge}} \cdot K_{\text{center}}},$$

where K_{center} is a target field exported to SYM-9 rather than silently set to the identity. The quotient removes gauge overlap; it does not erase the non-Abelian multiplicity-space action or the physical global currents that survive their quantum tests.

5.4 Abelian and anomaly ledger

The five Abelian directions can be expressed in many bases. Hypercharge, baryon number, total lepton number, family differences, and PQ-like complements are useful coordinates, but no Abelian basis has primitive authority. What matters is the current's action and anomaly vector.

ID	Current	Classical	Quantum / NP	Boundary
AB-01	U(1)_Y direction	gauge action	consistent only after Standard Model gauge- anomaly cancellation; gauge redundancy, not a global flavor charge	continuous direc- tion removed from faithful global fla- vor action
AB-02	B	accidental global symmetry at dimen- sion ≤ 4	mixed SU(2)_L anomaly; violated to- gether with L by elec- troweak topology	full operator- dimension analysis exported to SYM-7
AB-03	L	accidental global symmetry in the massless-neutrino renormalizable tar- get	mixed SU(2)_L anomaly; violated to- gether with B by elec- troweak topology	neutrino updates alter status
AB-04	B+L	accidental perturba- tive current	electroweak anoma- lous; violated by instan- ton/sphaleron sectors	SYM-7
AB-05	B-L	preserved by renor- malizable minimal SM and electroweak topology	mixed Standard Model gauge anomalies can- cel; preserved by elec- troweak topology	gauging requires field-content up- date
AB-06	L_i-L_j	exact in zero- neutrino-mass tar- get	gauge-anomaly-free fam- ily differences in the minimal local target; electroweak violation cancels between families	broken by generic neutrino mixing/- mass targets

ID	Current	Classical	Quantum / NP	Boundary
AB-07	PQ-like/e_R-like Abelian complements	basis-dependent linear combinations in the zero-Yukawa cover	generally anomalous and explicitly broken by Yukawa successors; combination-dependent	no privileged Abelian basis
AB-08	non-Abelian flavor background currents	zero-Yukawa limit action	nonzero flavor-hypercharge triangle data; higher-group restrictions depend on monopoles	requires magnetic one-form background and SYM-9 target

5.4.1 Baryon and lepton currents

In a left-handed Weyl convention the mixed $SU(2)_L^2 U(1)_B$ and $SU(2)_L^2 U(1)_L$ coefficients are nonzero, so B and L are not exact nonperturbative quantum symmetries. Electroweak topology violates $B + L$ while preserving $B - L$. The mixed Standard Model gauge anomalies of $B - L$ cancel, but without a right-handed neutrino its cubic and mixed-gravitational gauging tests do not vanish. Thus “ $B - L$ is anomaly free” is scope-sensitive: it is true for the local Standard Model gauge-anomaly entries relevant to perturbative conservation, not for every possible gauging/background question.

5.4.2 Family differences

With exactly massless neutrinos and no lepton-mixing tensor, $L_i - L_j$ currents survive the local gauge-anomaly and electroweak-topology tests because family contributions cancel. They cease to be exact physical symmetries in generic Dirac or Majorana mass targets. Their loss is a tensor deformation or target update, not a basis choice.

5.4.3 Non-Abelian flavor backgrounds

The non-Abelian factors are valid zero-Yukawa actions, yet their background-current triangle data with hypercharge enter the higher-flavor construction. The local statement and the higher-group statement are compatible: the latter requires one-form background and global-form data not present in the sharp-local target [12].

5.5 Faithful-qualification theorem

SYM5-T04 – Faithful quotient, Abelian, and anomaly qualification

The zero-Yukawa flavor record consists of the $U(3)^5$ local presentation cover, quotient by the continuous gauged hypercharge direction, a global-form-dependent discrete central kernel, and a current-by-current anomaly ledger. No classical $U(1)$ direction is promoted to exact quantum global status without its gauge, mixed, gravitational, and nonperturbative fields.

Proof / certificate

The commutant theorem supplies the local action. SYM-1 supplies the gauge representation and faithful-product boundary; SYM-3 supplies the distinct anomaly predicates. The machine-readable ledger contains separate rows for Y , B , L , $B + L$, $B - L$, family differences, complementary Abelian directions, and the higher-flavor background. The validator rejects an exact-quantum label on $B + L$ and requires the faithful-global-form field in the higher-flavor interface. The unresolved discrete center is therefore an explicit SYM-9 datum rather than a hidden premise of local closure.

Flavor action groupoids, basis quotients, and fixed-tensor stabilizers

6.1 The action groupoid

For a target $\mathcal{D}$, let $\mathcal{Y}_{\mathcal{D}}$ be the permitted tuple of Yukawa or mass tensors and $G_{\mathcal{D}}^{\text{basis}}$ the complete lawful weak-basis group. The action groupoid

$$\mathcal{G}_{\mathcal{D}} = \mathcal{Y}_{\mathcal{D}} // G_{\mathcal{D}}^{\text{basis}}$$

has tensor points as objects and basis transformations $g : y \rightarrow g \cdot y$ as arrows. The physical quotient is the orbit space

$$\mathcal{Q}_{\text{phys}}(\mathcal{D}) = \mathcal{Y}_{\mathcal{D}} / G_{\mathcal{D}}^{\text{basis}},$$

augmented by the discrete ordering, rephasing, anomaly, scale, and evidence data registered for the target.

The basic actions are

$$\begin{aligned} (Y_u, Y_d) &\mapsto (V_Q Y_u V_u^\dagger, V_Q Y_d V_d^\dagger), \\ (Y_e, Y_\nu) &\mapsto (V_L Y_e V_e^\dagger, V_L Y_\nu V_\nu^\dagger), \\ (Y_e, \kappa) &\mapsto (V_L Y_e V_e^\dagger, V_L^* \kappa V_L^\dagger). \end{aligned}$$

The last line is a congruence action on the complex symmetric Weinberg tensor, not a second Dirac biunitary action.

SYM5-T02 – Flavor action groupoid construction

For every closed target fiber, the registered field transformations define a compact group action on the target tensor space. The lawful arrows, kernels, quotient objects, successor maps, and source fields form a well-typed action groupoid and a versioned physical quotient.

Proof / certificate

Closure of each action follows from unitary matrix multiplication; identities and inverses are inherited from the basis group. Tensor laws preserve matrix type: complex matrices remain complex matrices and a symmetric κ remains symmetric under congruence. The target registry prevents arrows from crossing field or operator targets. The basis-action registry gives four canonical actions; the target-transport registry handles all changes that are not groupoid arrows.

6.2 Basis descent

A weak-basis arrow changes representatives while preserving every invariant function. If

$$q_{\mathcal{D}} : \mathcal{Y}_{\mathcal{D}} \longrightarrow \mathcal{Q}_{\text{phys}}(\mathcal{D})$$

is the quotient map, then

$$q_{\mathcal{D}}(g \cdot y) = q_{\mathcal{D}}(y) \quad (g \in G_{\mathcal{D}}^{\text{basis}}).$$

This equality is the physical content of basis equivalence. It does not say that g is a symmetry of the fixed numerical point y .

SYM5-T05 – Basis-quotient descent theorem

Every lawful weak-basis arrow descends trivially on the physical quotient. Conversely, a transformation that changes a registered spectrum, mixing invariant, CP invariant, anomaly field, operator content, or target fingerprint is not a same-target basis arrow.

Proof / certificate

Polynomial and spectral invariants are constant on group orbits. The quotient definition identifies precisely the orbit, together with discrete and evidence fields. The converse is project-relative: the candidate-generation protocol includes every standing-bearing field used by the paper, so changing one of them fails the groupoid-arrow or target-fingerprint check. The validator separately rejects hidden target changes and representative-dependent fake invariants.

6.3 The actual symmetry of a fixed tensor point

For $y \in \mathcal{Y}_{\mathcal{D}}$ define

$$\text{Stab}_{\mathcal{D}}(y) = \{g \in G_{\mathcal{D}}^{\text{basis}} : g \cdot y = y\}.$$

This is the physical flavor symmetry of the fixed coupling point. The full group $G_{\mathcal{D}}^{\text{basis}}$ instead acts on the family of tensor representatives. The distinction is exact:

$$\text{covariance of } \mathcal{Y}_{\mathcal{D}} \neq \text{symmetry of } (\mathcal{D}, y).$$

SYM5-T06 – Fixed-tensor stabilizer theorem

At a fixed target and tensor point y , a basis transformation is a physical internal flavor symmetry of that point if and only if it lies in $\text{Stab}_{\mathcal{D}}(y)$. Transformations outside the stabilizer are arrows to different tensor representatives in the same orbit when the tensors transform, or to different coupling points when numerical couplings are held fixed.

Proof / certificate

A physical symmetry of the fixed-coupling action must leave every coupling tensor unchanged under the simultaneous field action. This is exactly the stabilizer equation. If the tensor is transformed as a spurion, the same group element can express formal covariance of the family of actions, but the fixed background value has moved. Therefore formal spurion covariance and fixed-point symmetry occupy distinct loci.

6.4 Orbit dimension and effective action

For a compact Lie group acting smoothly on a stratum of tensor space,

$$\dim \text{Orb}(y) = \dim G_{\mathcal{D}}^{\text{basis}} - \dim \text{Stab}_{\mathcal{D}}(y),$$

and hence

$$\dim \mathcal{Q}_{\text{phys}}^{(\lambda)} = \dim \mathcal{Y}_{\lambda} - \dim G_{\mathcal{D}}^{\text{basis}} + \dim \text{Stab}_{\mathcal{D}}(y).$$

The stabilizer term is essential. Omitting it is equivalent to assuming that the group acts freely, which fails in every flavor target because at least a common baryon or lepton phase survives generically, and fails more strongly on special strata.

6.5 Factorization isotropy is not full-pair symmetry

An SVD or Takagi factorization can have a large isotropy group when singular values vanish or coincide. That isotropy describes nonuniqueness of the factorization of one tensor. The stabilizer of a complete pair such as (Y_u, Y_d) or (Y_e, κ) is the intersection imposed by both tensors. Consequently,

$$U(2) \text{ or } O(2) \text{ factorization isotropy} \not\equiv U(2) \text{ or } O(2) \text{ symmetry of the full pair.}$$

The stratum registry reports both quantities separately.

Quark flavor quotient, parameter count, invariants, and singular strata

7.1 Generic tensor target

The generic quark target is

$$\mathcal{Y}_Q = \text{Mat}_n(\mathbb{C}) \oplus \text{Mat}_n(\mathbb{C}), \quad G_Q = U(n)_Q \times U(n)_u \times U(n)_d,$$

with action

$$(Y_u, Y_d) \mapsto (V_Q Y_u V_u^\dagger, V_Q Y_d V_d^\dagger).$$

For a generic full-rank nondegenerate pair, the common stabilizer is the baryon phase

$$(V_Q, V_u, V_d) = (e^{i\alpha} \mathbf{1}, e^{i\alpha} \mathbf{1}, e^{i\alpha} \mathbf{1}),$$

so $\dim \text{Stab} = 1$.

7.2 Generic parameter-count theorem

The real tensor dimension is $4n^2$ and $\dim G_Q = 3n^2$. Therefore

$$\dim \mathcal{Q}_{\text{phys}}^Q = 4n^2 - 3n^2 + 1 = n^2 + 1.$$

The familiar decomposition is recovered without choosing a chart:

$$n^2 + 1 = 2n + \frac{n(n-1)}{2} + \frac{(n-1)(n-2)}{2}.$$

The terms are, respectively, $2n$ masses, $n(n-1)/2$ real mixing angles, and $(n-1)(n-2)/2$ CP phases.

SYM5-T08 – Quark and Dirac parameter-count theorem

For a generic n -generation pair of Dirac-type Yukawa matrices, the physical quotient has $n^2 + 1$ real parameters. At $n = 3$ the quark fiber contains six masses, three mixing angles, and one CP phase.

Proof / certificate

The dimension calculation above proves the quotient count. The decomposition follows by diagonalizing each matrix biunitarily and quotienting the relative left-handed unitary matrix by $2n - 1$ independent row/column phases. A unitary $n \times n$ matrix has n^2 real parameters, leaving $(n - 1)^2$ physical mixing parameters; these split into $n(n - 1)/2$ angles and $(n - 1)(n - 2)/2$ phases. Adding $2n$ singular values gives $n^2 + 1$.

7.3 Invariant reconstruction

Set

$$H_u = Y_u Y_u^\dagger, \quad H_d = Y_d Y_d^\dagger.$$

Both transform by simultaneous conjugation under $U(n)_Q$; right-handed basis factors disappear. For three generations, a convenient finite generic package contains

$$\text{tr}(H_u^k), \text{tr}(H_d^k), \quad k = 1, 2, 3,$$

for the spectra, together with mixed traces such as

$$\text{tr}(H_u H_d), \quad \text{tr}(H_u^2 H_d), \quad \text{tr}(H_u H_d^2), \quad \text{tr}(H_u^2 H_d^2),$$

and one CP-odd secondary. The invariant ring, its relations, and Hilbert-series count are imported at their exact target scope [6, 7]. The paper uses these invariants as a finite separating package on the generic stratum, not as a claim that a single polynomial chart is globally nonsingular.

7.4 Quark CP

A generalized CP action exists when unitary matrices X_Q, X_u, X_d satisfy

$$X_Q^\dagger Y_u X_u = Y_u^*, \quad X_Q^\dagger Y_d X_d = Y_d^*.$$

For three generations the standard CP-odd commutator invariant is

$$I_{\text{CP}}^Q = \text{Im tr}[H_u, H_d]^3.$$

Up to normalization it equals the CKM Jarlskog residue multiplied by the products of same-charge mass-squared differences [4]. Its value is independent of the weak basis. A complex entry in Y_u, Y_d , or a CKM chart is not by itself a CP certificate.

7.5 Registered singular strata

The generic count does not transport unchanged when tensor dimension or stabilizer changes. Four quark strata are closed in this release:

Q-GEN. Full rank, nondegenerate, generic orientation: $\dim \mathcal{Q}_{\text{phys}} = 10$.

Q-U0. One up-type singular value vanishes: $34 - 27 + 2 = 9$. The physical content is five masses, three angles, and one CP phase. A single massless quark does not generically remove the CKM phase.

Q-U21. Two nonzero up-type singular values coincide: $33 - 27 + 1 = 7$. The physical content is five distinct masses and two real mixing angles; the CKM phase is unphysical.

Q-COMM. $[H_u, H_d] = 0$ with generic nondegenerate spectra: $30 - 27 + 3 = 6$. Only the six masses remain.

The $U(2)$ isotropy of the degenerate up-type SVD in Q-U21 is not the full pair stabilizer: generic Y_d breaks it, leaving only $U(1)_B$. What the degeneracy removes is one dimension of the tensor stratum and two dimensions of physical mixing data relative to Q-GEN.

7.6 Quark normal form

The quotient-stable quark normal form consists of

(six or stratum-reduced masses, relative left-handed orbit,
CP-odd orientation, stratum ID, source/evidence ID).

A CKM matrix is a chart for the relative orbit after diagonalization. Rephasings and permitted permutations change that chart without changing the normal form. The Carrier ARC supports the flavor-transition and CP/holonomy roles; SYM-5 identifies the quotient data that occupy them.

Lepton target bifurcation: massless, Dirac, and Majorana fibers

8.1 Minimal renormalizable target

The minimal Standard Model contains L_L , e_R , and Y_e , but no ν_R and no Weinberg operator. The tensor space and basis group are

$$\mathcal{Y}_{L0} = \text{Mat}_n(\mathbb{C}), \quad G_{L0} = U(n)_L \times U(n)_e.$$

For generic Y_e , $\dim \mathcal{Y} = 2n^2$, $\dim G = 2n^2$, and the stabilizer is $U(1)^n$, giving n physical charged-lepton masses. At $n = 3$ there are exactly three flavor parameters in this sector. Because the neutrinos are exactly massless and no neutrino mass tensor fixes a second left-handed basis, a PMNS matrix is not a physical object of this target.

No-PMNS target guard

A unitary matrix inserted between a charged-lepton basis and an arbitrarily chosen basis of three massless neutrinos is removable by the unbroken $U(3)$ neutrino rotation. It is representative data, not a physical mixing residue. Physical PMNS structure appears only after a Dirac or Majorana neutrino tensor is added.

8.2 Dirac-neutrino target

Adding three right-handed neutrinos and a Dirac Yukawa tensor gives

$$\mathcal{Y}_{LD} = \text{Mat}_n(\mathbb{C})^2, \quad G_{LD} = U(n)_L \times U(n)_e \times U(n)_\nu.$$

This is algebraically parallel to the quark pair. The generic stabilizer is total lepton number $U(1)_L$, so

$$\dim \mathcal{Q}_{\text{phys}}^{LD} = n^2 + 1.$$

At $n = 3$: six masses, three angles, and one Dirac CP phase. Generalized CP requires unitary X_L, X_e, X_ν satisfying the two conjugation equations. The PMNS representative is quotiented by charged-lepton and neutrino rephasings just as the CKM representative is quotiented by quark rephasings [5].

The registered Dirac strata mirror the quark calculations:

- LD-GEN: $36 - 27 + 1 = 10$;
- LD-N0: $34 - 27 + 2 = 9$, with five masses, three angles, and one Dirac phase;

- LD-N21: $33 - 27 + 1 = 7$, with five distinct masses and two real mixing angles.

A single massless Dirac neutrino does not generically eliminate the Dirac phase; a twofold nonzero degeneracy does.

8.3 Dimension-five Majorana target

The low-energy Majorana target adds the Weinberg operator [13]. Its coefficient κ is a complex symmetric matrix, so

$$\mathcal{Y}_{LM} = \text{Mat}_n(\mathbb{C}) \oplus \text{Sym}_n(\mathbb{C}), \quad G_{LM} = U(n)_L \times U(n)_e.$$

The real dimension is

$$2n^2 + n(n+1) = 3n^2 + n.$$

For a generic pair the continuous stabilizer is absent, giving

$$\dim \mathcal{Q}_{\text{phys}}^{LM} = 3n^2 + n - 2n^2 = n(n+1).$$

At $n = 3$ this is twelve parameters: six masses, three mixing angles, one Dirac-type phase, and two Majorana phases.

SYM5-T09 – Majorana EFT parameter-count theorem

The generic n -generation low-energy Majorana flavor quotient has $n(n+1)$ real parameters. The additional $(n-1)$ phases relative to the Dirac target arise because independent rephasing of Majorana mass eigenfields is not available.

Proof / certificate

A complex $n \times n$ charged-lepton matrix contributes $2n^2$ real dimensions and a complex symmetric matrix contributes $n(n+1)$. The basis group has $2n^2$ dimensions. A generic pair has only a discrete common stabilizer, so the quotient dimension is $n(n+1)$. The mixing matrix has $n(n-1)/2$ angles and $n(n-1)/2$ phases, of which $(n-1)(n-2)/2$ are Dirac-type and $n-1$ are Majorana-type.

8.4 Majorana generalized CP

In the convention used here, CP conservation is equivalent to the existence of unitary X_L, X_e and an allowed intrinsic phase η such that

$$X_L^\dagger Y_e X_e = Y_e^*, \quad X_L^T \kappa X_L = \eta \kappa^*.$$

The placement of η is convention dependent; the existence of a compatible antiunitary structure is not. Weak-basis invariant criteria are attached to explicit nondegenerate, massless, partially degenerate, or fully degenerate hypotheses [10, 11].

8.5 Majorana singular strata

The Takagi factorization of κ has orthogonal isotropy on a nonzero degenerate subspace and unitary isotropy on a null subspace. These are factorization symmetries, not automatically symmetries of the full pair (Y_e, κ) .

The registered low-energy strata are:

LM-GEN. $30 - 18 = 12$: six masses, three angles, three CP phases.

LM-M0. Rank-two κ : $28 - 18 = 10$. Five masses, three angles, and two CP phases; one Majorana phase is removed.

LM-M21. Two equal nonzero Majorana masses: $28 - 18 = 10$. Five spectral parameters and five mixing coordinates, including two CP phases in the cited convention.

LM-M3. Three equal nonzero Majorana masses: $25 - 18 = 7$. Four spectral parameters and three quotient mixing coordinates. A generic PMNS angle/phase chart is not privileged on this stratum.

8.6 Neutrino bifurcation theorem

SYM5-T17 – Neutrino target bifurcation

The massless-neutrino minimal Standard Model, the Dirac-neutrino neighboring target, and the dimension-five Majorana target have distinct field content, tensor laws, basis groups, stabilizers, lepton-number status, parameter counts, rephasing quotients, mixing residues, and CP conditions. No PMNS or lepton-number conclusion transports between them without a registered target map.

Proof / certificate

The target registry exhibits the changed fields and operators. The basis-action registry shows biunitary Dirac and congruence Majorana laws. The parameter-count registry gives 3, 10, and 12 generic three-generation parameters for the three fibers. The transport registry marks the addition of ν_R as a field-content update and the Weinberg operator as an EFT update. The validator rejects either transition if it is relabelled as a representative transport.

Spurion covariance and conditional Minimal Flavor Violation

9.1 Three distinct objects

Spurion language is useful only if three objects remain separate:

1. the **formal covariance group** acting on fields and symbolic tensors;
2. the **fixed background tensor value** that breaks the formal group to its stabilizer;
3. an optional **dynamical field** whose fluctuations would require new kinetic, potential, and state data.

A Yukawa matrix used as a spurion is the second object, represented with a transformation law derived from the first. It is not automatically the third.

9.2 Quark spurions

Under

$$G_q = SU(3)_Q \times SU(3)_u \times SU(3)_d,$$

use the convention

$$Y_u \mapsto V_Q Y_u V_u^\dagger, \quad Y_d \mapsto V_Q Y_d V_d^\dagger.$$

Thus Y_u and Y_d transform as $(3, \bar{3}, 1)$ and $(3, 1, \bar{3})$, respectively. Formal covariance lets one construct flavor-singlet effective operators from Standard Model fields, Y_u , Y_d , and their conjugates. When the spurions are set to their observed background values, the formal group is broken to the fixed-pair stabilizer.

The Minimal Flavor Violation hypothesis declares that these background values are the complete sources of quark flavor breaking at the selected EFT order [3]. This is a conditional target statement. It does not follow from the renormalizable Standard Model alone, and it does not predict the numerical values of the spurions.

9.3 Strict and generalized CP conventions in MFV

The original strict MFV formulation can include CP invariance of flavor-blind Wilson coefficients, so that the Yukawa backgrounds are also the complete CP-breaking sources. Later generalized

variants admit additional flavor-blind phases. SYM-5 records this as a target field rather than treating both conventions as one theorem. The quark MFV closure therefore has the form:

complete declared spurions + fixed EFT order
+ coefficient CP convention $\implies$ closed operator grammar.

9.4 Minimal-field-content lepton MFV

For the selected low-energy Majorana target, use

$$G_\ell = SU(3)_L \times SU(3)_e$$

with Y_e and the symmetric neutrino tensor g_ν or κ as the declared breaking sources. In the present index convention,

$$Y_e \mapsto V_L Y_e V_e^\dagger, \quad \kappa \mapsto V_L^* \kappa V_L^\dagger.$$

The symmetric representation may be denoted $\mathbf{6}$ or $\bar{\mathbf{6}}$ depending on whether upper or lower flavor indices are chosen; the transformation law, not the typography of the representation label, is authoritative.

The minimal-field-content construction keeps the scale Λ_{LN} that violates total lepton number distinct from the scale Λ_{LFV} controlling charged-lepton flavor violation [8]. An extended-field-content seesaw formulation contains additional heavy-neutrino tensors and a larger flavor group. It is a successor target, not another chart of the same fiber.

9.5 Spurion no-promotion theorem

SYM5-T14 – Spurion covariance and no-promotion

Assigning a transformation law to a Yukawa or mass tensor organizes breaking ancestry and EFT covariance. It does not make a numerical tensor entry a primitive identity, an independent symmetry action, or a dynamical field. Adding a dynamical flavor multiplet changes the target.

Proof / certificate

A fixed spurion value has no independent configuration space, kinetic term, conjugate momentum, or fluctuation algebra in the target. Its transformation is a bookkeeping rule that restores covariance of a family of Lagrangians. At a fixed value, the physical symmetry is the stabilizer of that value. The Carrier ARC independently blocks primitive promotion of raw matrix entries and phase coordinates. The atlas therefore assigns spurions the role *breaking/response tensor* or *EFT route*, never an unqualified primitive occupant.

9.6 Conditional MFV closures

SYM5-T15 – Quark MFV conditional closure

Within D5-MFV-Q, at the fixed operator truncation and under the hypothesis that Y_u and Y_d are the complete flavor-breaking sources, every admitted operator coefficient is either expressible as a flavor-covariant spurion polynomial/convergent function of the registered invariant combinations or is recorded as a failure of the MFV target.

SYM5-T16 – Minimal lepton MFV conditional closure

Within D5-MFV-L, at the fixed EFT order and under the minimal-field-content hypothesis, Y_e and κ generate the declared lepton-flavor operator grammar. A heavy-neutrino tensor, a different lepton-number source, or an extended seesaw flavor group creates a successor target.

Proof / certificate

The two statements are conditional by construction. The spurion registry names the complete declared breaking set, covariance group, transformation law, EFT boundary, CP field, and successor trigger. The role certificate includes separate positions for operator grammar, scale boundary, no-promotion, and non-MFV/seesaw successors. An operator requiring an unregistered breaking tensor cannot be absorbed by changing notation; it fails the target hypothesis.

9.7 What MFV does and does not close

MFV closes a symmetry-organized operator grammar, not the dynamics of Wilson coefficients. It can relate flavor structures across processes once a target and power counting are fixed. It does not determine the EFT scale, the numerical coefficients, or the observed Yukawa hierarchy. Those remain downstream physical data.

Invariant algebras, generalized CP, and completeness discipline

10.1 Invariant functions as quotient coordinates

A polynomial invariant I satisfies

$$I(g \cdot y) = I(y) \quad (g \in G_{\mathcal{D}}^{\text{basis}}).$$

For compact groups acting linearly on finite-dimensional tensor spaces, polynomial invariants form a finitely generated algebra and separate closed orbits. Flavor tensors lie in this setting. The practical burden is not finite generation alone; it is to identify which finite package is generating, separating, algebraically independent, CP complete, or valid only on a specified stratum.

10.2 Five meanings of completeness

The release uses the word *complete* only with one of these qualifiers:

Generating. Every polynomial invariant belongs to the algebra generated by the listed elements.

Separating. Distinct closed orbits in the declared target/stratum have different values of at least one listed invariant.

Parameter-complete. The package reconstructs the physical quotient coordinates modulo declared discrete ambiguities.

CP-complete. Simultaneous vanishing of the listed CP-odd conditions is necessary and sufficient for generalized CP under stated hypotheses.

Certificate-complete. Every invariant row required by the target and role ledger has a source, scope, and successor rule.

A package can satisfy one meaning and not another.

10.3 Generic quark invariant package

For $H_u = Y_u Y_u^\dagger$ and $H_d = Y_d Y_d^\dagger$, the six traces

$$\text{tr}(H_u^k), \quad \text{tr}(H_d^k), \quad k = 1, 2, 3,$$

recover the unordered spectra at three generations. A convenient mixed CP-even package is

$$\mathrm{tr}(H_u H_d), \quad \mathrm{tr}(H_u^2 H_d), \quad \mathrm{tr}(H_u H_d^2), \quad \mathrm{tr}(H_u^2 H_d^2).$$

Together with a CP-odd secondary such as $\mathrm{Im} \mathrm{tr}[H_u, H_d]^3$, these give a finite generic separating normal form, subject to the algebraic relation recorded in the invariant-ring literature [6, 7]. The paper does not need a preferred choice among equivalent generating sets; it needs a frozen source-qualified package.

10.4 Rephasing invariants

After diagonalization, CKM and PMNS matrices retain row/column rephasing equivalences. Products of matrix elements in which every row and column phase cancels are quotient functions. The classification of independent rephasing invariants and their relations is imported from Jenkins–Manohar [5]. Their significance is structural:

matrix element phase is chart data,

whereas

closed rephasing product is a quotient residue.

Majorana targets admit fewer rephasings, so additional phase-sensitive invariants survive.

10.5 Generalized CP as an existence problem

The correct target-independent pattern is not “all tensors are real in the current basis.” It is the existence of a compatible antiunitary action. For Dirac-type pairs this is expressed by unitary matrices on every field species; for the Majorana target it is expressed by a congruence law on κ . These equations are invariant under weak-basis change because the CP matrices transform with the basis.

SYM5-T12 – Generalized-CP existence

A target tensor point is CP conserving if and only if there exists a target-lawful compatible set of unitary maps taking every tensor to its complex conjugate under the appropriate biunitary or congruence law. Reality in one basis is a sufficient representative display when such a basis exists, not the definition of physical CP.

Proof / certificate

The “only if” direction extracts the unitary part of the antiunitary CP action on each flavor multiplicity space. The “if” direction composes complex conjugation with the unitary maps and verifies invariance of every target tensor. The equations are exactly those recorded in the CP registry. Weak-basis covariance follows by conjugating the X matrices with the basis transformations. The target-specific Majorana condition is imported from the cited weak-basis analyses [10, 11].

10.6 Target- and stratum-complete CP certificates

SYM5-T13 – Target- and stratum-complete CP certificate

Every CP conclusion in SYM-5 is accompanied by a registered generalized-CP equation and a necessary-and-sufficient invariant package at its declared target and stratum. A criterion may not be transported to a singular or Majorana stratum unless the source theorem and target map explicitly authorize that transport.

The registry implements this rule as follows:

- the quark and Dirac three-generation targets use the commutator/Jarlskog class at their declared scope;
- the nondegenerate Majorana target uses a source-qualified weak-basis invariant set;
- the massless, partial-degeneracy, and complete-degeneracy Majorana strata receive separate successor packages;
- a new orbit type is marked unclosed rather than filled by a generic criterion.

10.7 Representative/invariant non-collapse

SYM5-T11 – Representative/invariant non-collapse

Matrix entries, diagonalization matrices, eigenvector phases, CKM/PMNS parameterizations, and branch/order conventions are representative data. Singular values, relative eigenspace geometry modulo the target quotient, rephasing-invariant products, invariant polynomials, and target-qualified CP residues descend to physical quotient classes.

Proof / certificate

Every representative item changes under at least one lawful basis or chart transformation. Every listed invariant item is constant on those transformations by construction. The distinction is verified row by row in the invariant and flavor-atlas registries; the mutation suite rejects a chart-dependent phase inserted as an invariant.

10.8 Published invariant registry

ID	Target / str.	Invariant object	Completeness
INV-Q-10	D5-Q-GEN / Q-GEN	$\text{tr}(H_u^k), \text{tr}(H_d^k),$ $k=1,2,3$	generic and all strata as spectral data
INV-Q-11	D5-Q-GEN / Q-GEN	$\text{tr}(H_u H_d)$	part of finite generic sepa- rating package
INV-Q-12	D5-Q-GEN / Q-GEN	$\text{tr}(H_u^2 H_d)$	part of finite generic sepa- rating package
INV-Q-13	D5-Q-GEN / Q-GEN	$\text{tr}(H_u H_d^2)$	part of finite generic sepa- rating package

ID	Target / str.	Invariant object	Completeness
INV-Q-14	D5-Q-GEN / Q-GEN	$\text{tr}(H_u^2 H_d^2)$	part of finite generic separating package
INV-Q-CP	D5-Q-GEN / Q-GEN, Q-U0, Q-U21, Q-COMM	$\text{Im tr}([H_u, H_d]^3)$ (equivalently the three-generation Jarlskog commutator determinant up to normalization)	three-generation quark CP criterion; interpretation is stratum-sensitive
INV-LD-CP	D5-L-DIRAC / LD-GEN, LD-N0, LD-N21	Dirac analogue $\text{Im tr}([H_e, H_\nu]^3)$	three-generation Dirac target with target hypotheses
INV-LM-WB	D5-L-MAJ5 / LM-GEN	registered weak-basis invariants I_1, I_2, I_4 (source convention)	hypothesis-indexed; not transported to degeneracy
INV-LM-21	D5-L-MAJ5 / LM-M21	partial-degeneracy invariant pair/package in source convention	necessary/sufficient under cited stratum assumptions
INV-LM-M0	D5-L-MAJ5 / LM-M0	massless-neutrino successor invariant package	stratum-specific
INV-LM-M3	D5-L-MAJ5 / LM-M3	complete-degeneracy invariant package	stratum-specific; no generic chart transport

Orbit-type strata, rank loss, and degeneracy stabilizers

11.1 Why generic formulas are insufficient

A rank or degeneracy condition changes at least one of:

- the dimension of the tensor stratum;
- the isotropy of an SVD or Takagi factorization;
- the stabilizer of the complete tensor tuple;
- the dimension and topology of the physical quotient;
- the number and organization of mixing/CP coordinates;
- the invariant package required for orbit separation or CP completeness.

Accordingly, degeneracy is not a footnote to a generic chart. It is an orbit-type target field.

11.2 Dimension formula for a complex matrix stratum

Let an $n \times n$ complex matrix have positive singular-value multiplicities $m_1, \dots, m_s$ and nullity k . Under $U(n)_L \times U(n)_R$, the factorization isotropy has real dimension

$$\sum_{a=1}^s m_a^2 + 2k^2.$$

The stratum dimension is

$$2n^2 - \sum_{a=1}^s m_a^2 - 2k^2 + s.$$

The final s counts the distinct positive singular values. This formula computes the one-matrix stratum. The full-pair stabilizer must still be obtained by intersecting the conditions imposed by the second tensor.

11.3 Dimension formula for a complex symmetric stratum

For a complex symmetric matrix under unitary congruence, positive Takagi multiplicities m_a have orthogonal isotropy and the null subspace has unitary isotropy. The factorization isotropy

dimension is

$$\sum_{a=1}^s \frac{m_a(m_a - 1)}{2} + k^2,$$

and the stratum dimension is

$$n^2 - \sum_{a=1}^s \frac{m_a(m_a - 1)}{2} - k^2 + s.$$

At $n = 3$, a generic full-rank symmetric matrix has dimension 12, a rank-two generic symmetric matrix has dimension 10, a $2 + 1$ nonzero degeneracy also has dimension 10, and a fully degenerate nonzero matrix has dimension 7.

11.4 Registered stratum table

Stratum	Target	dim T	dim G	dim Q	Physical decomposition
Q-GEN	D5-Q-GEN	36	27	10	6 masses + 3 angles + 1 CP phase
Q-U0	D5-Q-STRAT	34	27	9	5 masses + 3 angles + 1 CP phase
Q-U21	D5-Q-STRAT	33	27	7	5 distinct masses + 2 real mixing angles; no physical CKM phase
Q-COMM	D5-Q-STRAT	30	27	6	6 masses; no physical mixing
L0-GEN	D5-L-0NU	18	18	3	3 charged-lepton masses; no physical PMNS data
LD-GEN	D5-L-DIRAC	36	27	10	6 masses + 3 angles + 1 Dirac phase
LD-N0	D5-L-DIRAC	34	27	9	5 masses + 3 angles + 1 Dirac phase
LD-N21	D5-L-DIRAC	33	27	7	5 distinct masses + 2 real mixing angles; no Dirac phase
LM-GEN	D5-L-MAJ5	30	18	12	6 masses + 3 angles + 3 CP phases
LM-M0	D5-L-MAJ5	28	18	10	5 masses + 3 angles + 2 CP phases
LM-M21	D5-L-MAJ5	28	18	10	5 distinct masses + 5 mixing coordinates, including 2 CP phases under the cited convention
LM-M3	D5-L-MAJ5	25	18	7	4 spectral parameters + 3 quotient mixing coordinates; no generic PMNS chart is privileged

11.5 Full-pair stabilizer versus factorization isotropy

For Q-U21, the degenerate Y_u factorization admits $U(2)$ rotations, but a generic Y_d breaks them; the full pair retains only baryon number. For LM-M21, the Takagi factorization of κ admits $O(2)$, but a generic Y_e need not commute with that rotation; the full pair has no continuous stabilizer. These examples show why an isotropy group read from one diagonal matrix cannot be inserted directly into the physical quotient dimension of a tensor pair.

11.6 Phase survival

The registered counts give the following exact lessons.

1. One massless quark or one massless Dirac neutrino adds a right-handed null-space phase to the stabilizer but leaves the single three-generation Dirac CP phase physical in a generic pair.
2. A twofold nonzero same-charge Dirac degeneracy removes the physical Dirac phase and one mixing angle relative to the generic pair.
3. One massless Majorana neutrino removes one Majorana phase but not the Dirac-type phase in a generic mixing configuration.
4. A partial Majorana degeneracy reorganizes the quotient and leaves two CP phases in the cited low-energy convention [11].
5. Complete Majorana degeneracy leaves a nontrivial quotient between the charged-lepton tensor and the degenerate Majorana form; it is not equivalent to “no mixing.”

SYM5-T07 / T18 – Registered orbit-type and neutrino stabilizer closure

Every singular or degenerate stratum used by a parameter, mixing, or CP claim has an explicit tensor condition, tensor dimension, basis-group dimension, full-pair stabilizer dimension, quotient dimension, factorization isotropy, physical decomposition, CP package, and successor status. The twelve registered rows satisfy

$$\dim \mathcal{Q}_{\text{phys}}^{(\lambda)} = \dim \mathcal{Y}_\lambda - \dim G^{\text{basis}} + \dim \text{Stab}_{\text{pair}}.$$

Proof / certificate

The dimension identities are checked executably for all twelve rows. The complex-matrix and complex-symmetric formulas establish the tensor-stratum dimensions. Direct stabilizer equations establish the full-pair entries. The invariant registry then assigns target/stratum-qualified CP packages. The validator rejects an invariant or CP row naming an unregistered stratum.

11.7 Closure boundary

The theorem is complete for the twelve registered strata used by the paper. It does not assert that these are every orbit type of every arbitrary flavor tensor model. A new same-scope stratum in one of the frozen tensor spaces creates a versioned registry successor and must receive the same dimension, stabilizer, invariant, and CP fields before transport.

Higher-flavor interface and the global-form boundary

12.1 Why local $U(3)^5$ is not the final generalized-symmetry record

A local current algebra and a defect-completed global symmetry are different theorem objects. The higher-flavor construction couples zero-form flavor backgrounds to the magnetic hypercharge one-form symmetry through nonzero triangle data [12]. To formulate that statement, the target must specify the faithful Standard Model gauge group, genuine line spectrum, magnetic background, monopole assumptions, and allowed defects.

12.2 Interface data

The interface registry requires seven fields:

1. faithful global gauge form;
2. magnetic hypercharge one-form background;
3. genuine line/defect operator category;
4. zero-form flavor background fields;
5. mixed flavor–hypercharge triangle coefficients;
6. higher-group extension/Postnikov data;
7. monopole and defect assumptions.

Omitting any of these fields changes the meaning of the claim. In particular, dynamical monopoles can remove or modify the magnetic one-form symmetry that participates in the higher group.

12.3 Conditional interface theorem

SYM5-T19 – Higher-flavor conditional interface

If a target fixes the faithful Standard Model global form, genuine line/defect spectrum, magnetic hypercharge one-form background, flavor backgrounds, triangle coefficients, and monopole assumptions, then the imported higher-flavor construction determines a zero-form/one-form higher-group extension and a corresponding Yukawa-spurion transformation rule. Without

those data, SYM-5 exports the local current/anomaly information but does not assert defect-completed higher-group closure.

Proof / certificate

Córdova–Koren supply the primary QFT implication from the mixed triangle data to the higher-group structure under the stated background assumptions. SYM-3 supplies the anomaly type and conservative transport rule. The interface registry encodes every required premise. The validator rejects the target if the faithful global form or line category is removed. Therefore the local theorem and the generalized-symmetry theorem are connected by an exact conditional map rather than conflated.

12.4 Yukawas as higher spurions

In the extended target, Yukawa tensors that break the zero-form flavor action can be assigned transformation data compatible with the higher group. This does not alter their local no-promotion status. It enriches the background covariance and UV-completion constraints; it does not make the numerical Yukawa entries dynamical or primitive.

12.5 SYM-9 handoff

SYM-5 exports:

- the local zero-Yukawa flavor presentation and current anomaly fields;
- the faithful-quotient placeholder and global-form dependence;
- the required one-form background and triangle data;
- the conditional higher-group extension field;
- the monopole, line, and defect burdens that prevent local overclaiming.

SYM-9 owns the complete global-form family, line spectrum, one-form symmetry, theta periodicity, monopole sector, and defect-completed closure. This division is compositional: no SYM-5 theorem must be withdrawn when SYM-9 supplies the missing global data.

AASC role map, no-promotion results, and integrated normal form

13.1 Role vocabulary

After the framework-neutral QFT ledger is fixed, every object is assigned one or more roles at explicit loci:

Limit physical action. A nontrivial action in the zero-Yukawa or parameter-limit target.

Fixed-point action. The stabilizer of a specific tensor point.

Basis skin. A transformation or chart change that preserves the complete physical quotient.

Breaking/response tensor. A Yukawa or mass tensor whose value changes spectra, mixing, or symmetry.

Spurion/EFT route. Formal covariance used to organize a neighboring EFT target.

Invariant witness. A spectrum, trace, determinant, rephasing product, CP invariant, or dimension certificate.

Obstruction. An anomaly, lepton-number-breaking term, or failed compatibility condition.

Interface. A typed handoff requiring another target's global data.

Target update. A change of fields, operators, stratum, global form, scale, or evidence identity.

Multi-role assignment is allowed only across distinct loci. A basis group can be a formal covariance route while its fixed-point stabilizer is a physical action and its generic arrows are skin.

13.2 Representative/tensor/action distinctions

The central noncollapse results are:

$$\begin{aligned}
 G^{\text{basis}} \text{ arrow} &\neq \text{Stab}(y) \text{ action}, \\
 \text{spurion law} &\neq \text{dynamical flavor field}, \\
 \text{matrix phase} &\neq \text{CP invariant}, \\
 \text{factorization isotropy} &\neq \text{full-pair stabilizer},
 \end{aligned}$$

and

parameter-limit symmetry $\neq$ generic-target symmetry.

Each inequality is backed by a quotient, stabilizer, target, or source certificate rather than terminological preference.

13.3 Carrier support without authority inflation

The Carrier ARC identifies primitive support in the flavor-transition and CP/holonomy-residue package. SYM-5 realizes that support through quotient-stable objects:

- singular-value spectra;
- relative left-handed eigenspace orbit;
- rephasing and permutation quotient;
- target/stratum-qualified CP residue;
- breaking ancestry and target transport.

Raw CKM/PMNS entries, basis vectors, ordering labels, and phase conventions remain representative. They can be calculationally indispensable without becoming primitive identities.

13.4 Complete locus-sensitive role theorem

SYM5-T20 – Complete locus-sensitive flavor role classification

Every registered flavor object occupies exactly the roles licensed by its target, quotient action, fixed-point behavior, invariant content, anomaly field, and successor status. No object is classified as both skin and an unqualified physical action at the same locus; no raw matrix entry or spurion symbol receives primitive standing.

Proof / certificate

The flavor atlas contains 82 records paired to 82 QFT residues. Basis actions and charts are marked skin; fixed-point stabilizers and conserved limit currents are actions; Yukawa and Majorana tensors are tensor/obstruction data; invariant polynomials and parameter counts are witnesses; MFV grammars are routes; higher flavor is an interface; and neutrino/global-form changes are target updates. The validator enforces target, kind, source, anomaly, and interface consistency. The no-promotion mutation test rejects a representative-dependent fake invariant.

13.5 Kernel necessity and positive construction

The QFT construction does not float free of determinacy: its targets, arrows, equivalence classes, invariants, and records require admissibility, standing, reference, and fixation to be determinate objects. But the kernel does not choose $U(3)^5$, compute a Hilbert series, or decide whether the neutrino tensor is Dirac or Majorana. Those positive results remain the responsibility of QFT, mathematics, and the cited sources. This preserves the locked dependency order without turning AASC into an optional overlay.

13.6 Integrated normal form

For each closed target/stratum, the final normal form is

$$\mathbf{NF}_{\mathcal{D},\lambda} = (\mathfrak{f}(\mathcal{D}), [y]_{G^{\text{basis}}}, \text{Stab}(y), \text{Inv}_{\mathcal{D},\lambda}(y), \text{CP}_{\mathcal{D},\lambda}, \text{Anom}_{\mathcal{D}}, \text{Role}_{\mathcal{D},\lambda}, E).$$

It preserves target identity, quotient class, full-pair stabilizer, invariant package, CP status, anomaly vector, role vector, and evidence version. Any presentation that preserves this tuple is skin; any change to a component is tensor/action data or a target successor.

SYM5-T23 – Integrated flavor normal form

The target, groupoid, stabilizer, stratum, invariant, CP, anomaly, spurion, role, and evidence registries compose into $\mathbf{NF}_{\mathcal{D},\lambda}$ for every closed fiber. The normal form is invariant under lawful weak-basis redescription and conservative under registered successor transport.

Finite role-occupation certificate

14.1 Purpose and theorem boundary

The finite certificate verifies that the framework-neutral QFT objects and the AASC role burdens form one consistent classification at the frozen target scope. It does not prove the imported QFT theorems by enumeration. Those theorems supply the residue fields; the certificate proves coverage, nonduplication, compatibility, relation preservation, and uniqueness of the role assignment modulo lawful automorphisms.

14.2 Independent constructor surfaces

Two canonical inputs are maintained:

- `constructors/role_burdens.json`: 82 positions formed in burden language from the theorem and role obligations;
- `constructors/qft_residues.json`: 82 framework-neutral objects formed in QFT language with source/import identifiers.

They use disjoint `ROLE-` and `QFT-` namespaces, distinct files and hashes, and no expected-assignment field. Their relation graphs are also stored separately. Both constructors necessarily describe the same frozen targets; independence means that neither input contains the other's node IDs or a prewritten map.

Independence claim

The certificate establishes *construction-surface independence*: separate authorship basis, files, hashes, namespaces, relation ledgers, and absence of an expected map. It does not claim that the two descriptions are epistemically unrelated; both are descriptions of the same flavor architecture and therefore share target and interface semantics.

14.3 Target blocks and complete pair evaluation

The block sizes are

$$5, 7, 9, 9, 10, 10, 10, 11, 11,$$

which sum to 82. Every role is compared with every residue in the same target block, giving

$$5^2 + 7^2 + 9^2 + 9^2 + 10^2 + 10^2 + 10^2 + 11^2 + 11^2 = 778$$

evaluated pairs. No cross-target pair is eligible because target identity is standing-bearing data. Each pair receives three primary checks:

1. kind compatibility between the burden and the QFT object;
2. equality of the independently stated semantic interface signature;
3. presence of a source or local-derivation certificate.

Every role has exactly one eligible residue; every residue is used exactly once.

14.4 Relational compatibility

Nodewise eligibility is necessary but not the only audit. Each target carries a directed burden relation graph and a separately authored QFT dependency graph. Under the candidate assignment, every burden edge must map to a QFT dependency edge. The canonical assignment preserves all edges; the relation-failure ledger is empty.

A second implementation uses `networkx.DiGraphMatcher` with node colors given by target, kind, and semantic signature. Every target block has exactly one colored directed graph isomorphism, and it equals the canonical assignment. The product of blockwise counts is one.

14.5 Automorphism quotient

A lawful automorphism must preserve target, kind, semantic signature, evidence requirement, and relation graph. Every semantic signature is unique within its target block, so the only lawful automorphism is the identity. Hence the single complete assignment forms one orbit:

$$|\text{Assign}| = 1, \quad |\Gamma^{\text{lawful}}| = 1, \quad |\text{Assign} / \Gamma^{\text{lawful}}| = 1.$$

The label-invariance mutation test then renames every opaque role and residue ID while preserving the semantic and relation fields; the certificate still passes. Uniqueness therefore does not depend on the original identifier strings.

14.6 Certificate result

Executable certificate result	
Role positions	82
QFT residues	82
Same-target pairs evaluated	778
Eligible pairs	82
Complete compatible assignments	1
Lawful automorphisms	1
Lawful assignment orbits	1
Independent NetworkX replay	PASS, one isomorphism in every target block
Mutation tests	12/12 PASS
Validation errors / relation failures	0 / 0

SYM5-T21 – Independent role-position and residue construction

The role positions and QFT residues are independently represented at the canonical input surface, with disjoint IDs, separate source paths and hashes, no expected map, and separately authored relation graphs.

SYM5-T22 – Fiberwise unique assignment-orbit theorem

For every registered target block, all same-target pairs are evaluated, exactly one complete relation-preserving assignment survives, and the lawful automorphism quotient contains one orbit. The product over all nine blocks also contains one orbit.

Proof / certificate

The canonical validator proves pair completeness, unique eligibility, bijectivity, evidence presence, relation preservation, and certificate self-consistency. The independent NetworkX replay proves one colored directed graph isomorphism in each block. The mutation suite shows that target corruption, missing roles, kind changes, relation deletion, duplicate IDs, anomaly laundering, stratum overreach, counterfeit invariants, missing evidence, incomplete higher-flavor data, and hidden neutrino updates are rejected, while opaque label permutation is accepted.

14.7 Interpretive limit

Because interface signatures are deliberately precise, the finite search is a consistency and uniqueness certificate for a completed classification, not an unsupervised discovery procedure. Its scientific value is that every role claim is exposed as a machine-readable burden, every QFT object is independently sourced, all alternatives in the frozen blocks are evaluated, and the surviving map is replayable.

Integrated theorem ladder and end-point proof

15.1 Status vocabulary

The theorem registry separates:

Proved. A local definition, dimension calculation, quotient theorem, transport theorem, or composition theorem discharged in this project.

Imported and specialized. A primary QFT result specialized to the frozen target and supplied with a local scope/translation proof.

Conditional closure proved. A theorem proved under an explicit MFV or interface hypothesis.

Finite certificate passed. A complete executable enumeration and independent replay.

Proved at declared scope. A capstone whose exclusions are part of the quantifier.

No theorem is promoted merely because the same statement occurs in multiple corpus documents.

15.2 Registry

ID	Result	Status
SYM5-T01	Target-family and fingerprint completeness	PROVED
SYM5-T02	Flavor action groupoid construction	PROVED
SYM5-T03	Zero-Yukawa commutant theorem	IMPORTED AND SPECIALIZED
SYM5-T04	Faithful quotient, Abelian and anomaly qualification	PROVED WITH GLOBAL FORM BOUNDARY
SYM5-T05	Basis-quotient descent theorem	PROVED
SYM5-T06	Fixed-tensor stabilizer theorem	PROVED
SYM5-T07	Orbit-type and degeneracy stratification	PROVED FOR REGISTERED STRATA
SYM5-T08	Quark and Dirac parameter-count theorem	PROVED

ID	Result	Status
SYM5-T09	Majorana EFT parameter-count theorem	PROVED
SYM5-T10	Finite invariant algebra and orbit-separation theorem	IMPORTED AND SPECIALIZED
SYM5-T11	Representative/invariant non-collapse theorem	PROVED
SYM5-T12	Generalized-CP existence theorem	IMPORTED AND SPECIALIZED
SYM5-T13	Target- and stratum-complete CP certificate	CERTIFIED BY TARGET STRATUM REGISTRY
SYM5-T14	Spurion covariance and no-promotion theorem	PROVED
SYM5-T15	Quark MFV conditional closure	CONDITIONAL CLOSURE PROVED
SYM5-T16	Minimal lepton MFV conditional closure	CONDITIONAL CLOSURE PROVED
SYM5-T17	Neutrino target bifurcation theorem	PROVED
SYM5-T18	Massless and degenerate neutrino stabilizer theorem	PROVED FOR REGISTERED STRATA
SYM5-T19	Higher-flavor conditional interface	CONDITIONAL INTERFACE PROVED
SYM5-T20	Complete locus-sensitive flavor role classification	PROVED
SYM5-T21	Independent role-position and residue construction	PROVED
SYM5-T22	Fiberwise unique assignment-orbit theorem	FINITE CERTIFICATE PASSED
SYM5-T23	Integrated flavor normal-form theorem	PROVED
SYM5-T24	Conservative successor and downstream transport	PROVED
SYM5-T25	Integrated flavor endpoint closure	PROVED AT DECLARED SHARP LOCAL SCOPE

15.3 Foundation layer: T01–T07

T01 freezes the target family and successor discipline. **T02** constructs the action groupoids. **T03** identifies the $U(3)^5$ local commutant. **T04** applies the faithful gauge-overlap and anomaly qualifications. **T05** proves basis descent. **T06** identifies physical fixed-point symmetry with the stabilizer. **T07** closes the twelve registered orbit strata.

The load-bearing implication is

$$\text{fixed representation and tensor law} \implies \mathcal{Y}_{\mathcal{D}} // G_{\mathcal{D}}^{\text{basis}} \implies ([y], \text{Stab}(y), \lambda(y)).$$

Neither $[y]$ nor $\text{Stab}(y)$ can be read from the group name alone.

15.4 Physical-data layer: T08–T13

T08 and **T09** derive generic Dirac and Majorana parameter counts. **T10** imports finite invariant algebras at exact target scope. **T11** separates representatives from quotient data. **T12** states generalized CP as the existence of compatible unitary maps. **T13** attaches necessary-and-sufficient CP packages to target/stratum hypotheses.

Their composition yields

$$[y]_{G^{\text{basis}}} \longmapsto (\text{spectra, relative orbit, CP residue, } \lambda),$$

with a versioned successor whenever the generic package ceases to be complete.

15.5 EFT and frontier layer: T14–T19

T14 proves spurion no-promotion. **T15** and **T16** close quark and minimal-lepton MFV conditionally. **T17** proves neutrino target bifurcation. **T18** closes registered massless/degenerate neutrino strata. **T19** proves the exact conditional higher-flavor handoff.

The common pattern is

$$\text{new tensor, field, operator, scale, or background} \implies \text{successor target},$$

not a silent reinterpretation of the original quotient.

15.6 Role and certificate layer: T20–T24

T20 gives the locus-sensitive role map. **T21** establishes separate constructor surfaces. **T22** proves one assignment orbit. **T23** composes the integrated normal form. **T24** proves conservative transport: unchanged target and evidence fields transport unchanged; changed fields create successors.

15.7 Endpoint theorem

SYM5-T25 – Kernel-Faithful Integrated Flavor Action–Quotient–Spurion–Invariant Occupancy Closure

Fix the nine-target registry, twelve registered orbit strata, completed upstream interfaces, primary-source map, and frozen candidate-generation protocol. For every sharp-local or declared conditional SYM-5 fiber:

1. the framework-neutral flavor action, basis quotient, fixed-tensor stabilizer, invariant/CP package, anomaly vector, spurion status, and successor boundary are completely typed;
2. role assignment descends to the physical quotient and is stable under lawful redescription;
3. the 82 role positions and 82 QFT residues admit one complete lawful assignment orbit after all 778 same-target pairs and relation constraints are evaluated;
4. every registered rival is rejected, absorbed, or redirected without leaving a genuine same-scope counterexample.

Therefore no untyped same-scope flavor role or governance-inequivalent second classifier survives within the declared sharp-local SYM-5 scope. The higher-flavor defect completion and whole-arc

exhaustion are excluded from the quantifier and exported to SYM-9 and SYM-11, respectively.

Proof / certificate

T01–T07 establish target, action, quotient, stabilizer, and stratum well-formedness. T08–T13 establish quotient dimensions, invariant data, representative descent, and CP completeness at the declared hypotheses. T14–T19 establish spurion/MFV status, neutrino target separation, singular lepton strata, and the higher-flavor boundary. T20 classifies every registered object by locus. T21 and T22 prove independent-constructor surface and one assignment orbit. T23 composes the normal form; T24 guarantees conservative successor transport. The source, validation, NetworkX replay, mutation, rival, and nonclaim ledgers all pass, and the genuine-counterexample ledger is empty. These facts establish the stated finite-scope conclusion.

15.8 Why the theorem is not circular

The QFT target and residues are fixed before AASC roles are assigned. The role constructor uses theorem burdens; the QFT constructor uses actions, tensors, invariants, anomalies, and sources. Compatibility is tested only after both exist. The kernel explains why determinate targets and quotient-stable standing are necessary; it does not appear as a premise in the commutant, parameter-count, invariant-ring, or CP calculations.

Atlas, software, and reproducibility architecture

16.1 Controlled-object principle

The machine-readable target, theorem, source, group, anomaly, stratum, invariant, CP, spurion, role, residue, eligibility, assignment, rival, nonclaim, and mutation registries are primary project objects. The manuscript is a versioned projection of them. No load-bearing flavor status exists only in prose.

16.2 Repository layout

```
SYM5_Full_Audit_Grade_v1.0.0/
  main.tex, references.bib, requirements.txt, sections/, appendices/,
  generated/
  data/
    target_registry.{csv,json}
    theorem_registry.{csv,json}
    limit_group_ledger.{csv,json}
    abelian_anomaly_ledger.{csv,json}
    basis_action_registry.{csv,json}
    parameter_count_registry.{csv,json}
    stratum_registry.{csv,json}
    invariant_registry.{csv,json}
    cp_registry.{csv,json}
    spurion_registry.{csv,json}
    higher_flavor_interface.{csv,json}
    flavor_atlas.{csv,json}
    eligibility_matrix.{csv,json}
    canonical_assignment.{csv,json}
    assignment_certificate.{csv,json}
    mutation_results.{csv,json}
  constructors/
    role_burdens.{csv,json}, role_relations.{csv,json}
    qft_residues.{csv,json}, qft_relations.{csv,json}
  tools/
    generate_release_data.py
    validate_release.py
    replay_certificate.py
    run_mutation_tests.py
    pdf_preflight.py
    hash_manifest.py
```

audit/, governance/, schemas/, build/

16.3 Validator rules

The release validator rejects:

- missing, duplicate, or unknown target, theorem, role, residue, source, stratum, or interface identifiers;
- a basis-equivalence row marked physical without a distinct stabilizer locus;
- a CP conclusion without a generalized-CP or source-qualified invariant certificate;
- a generic invariant package applied to an unregistered singular stratum;
- an anomalous current marked exact quantum conservation;
- a higher-group row lacking faithful global form, line category, or magnetic background;
- a primitive role assigned to a raw matrix entry, chart phase, or spurion symbol;
- a target change hidden as a representative transformation;
- an incomplete all-pairs matrix, non-bijective assignment, missing relation edge, or constructor hash mismatch;
- any theorem registry entry retaining a planned or skeleton status.

16.4 Executable checks

The build runs, in order:

1. deterministic regeneration of all canonical ledgers and generated TeX tables;
2. schema and cross-reference validation;
3. independent NetworkX graph-isomorphism replay;
4. twelve mutation tests;
5. LaTeX and bibliography compilation;
6. undefined-reference, undefined-citation, overfull-box, and font checks;
7. PDF structural preflight and full-page rendering;
8. project-wide SHA-256 manifest generation;
9. clean source/PDF packaging.

16.5 Artifact invariants

The release freezes the following counts:

(9 targets, 25 results, 12 strata, 82 roles, 82 residues, 778 pairs).

A change to any count requires a versioned successor unless it is a packaging-only change that leaves all canonical records and theorem identities fixed.

16.6 Formalization boundary

The executable certificate formalizes finite record consistency and graph matching, not the interacting Standard Model. A future Lean layer may encode the target datatype, well-formedness predicates, finite eligibility relation, automorphism action, and singleton-orbit theorem. Imported QFT theorems remain named interfaces unless their finite-dimensional algebraic specializations are separately mechanized.

16.7 Rebuild command

From a clean project directory, the complete release is rebuilt with

```
./build.sh
```

The script exits nonzero on a data, certificate, mutation, LaTeX, citation, or preflight failure. The build directory contains the final PDF, audit logs, PDF metadata/font reports, and manifest.

Adversarial map and rival disposition

17.1 Standing requirement

A hostile objection has standing only if it preserves the declared target and identifies a failed QFT theorem, quotient map, stabilizer calculation, invariant-completeness claim, anomaly field, target transport, constructor-independence condition, eligibility/compatibility edge, automorphism quotient, or source certificate. Restating a blocked inference without tracing its theorem chain is not a surviving objection.

17.2 Frozen rival ledger

ID	Rival claim	Disposition	Failed implication
RV-01	The full weak-basis group is a physical symmetry of a fixed numerical Yukawa target	REJECTED	covariance group => fixed-point stabilizer
RV-02	Every $U(1)$ factor in $U(3)^5$ is an exact quantum global symmetry	SPLIT AND REJECTED	classical commutant => exact quantum current
RV-03	A chosen complex Yukawa phase is physical	REJECTED AS SKIN	complex representative => quotient CP residue
RV-04	One generic invariant package is complete on every singular and Majorana stratum	REJECTED	generic separation => stratum-universal separation
RV-05	Spurions are new dynamical primitive fields	REJECTED	formal covariance => dynamical/primitive status
RV-06	MFV is forced by the Standard Model	TARGET UPDATE	useful EFT hypothesis => unconditional target theorem
RV-07	Dirac and Majorana PMNS parameterizations are the same target	REJECTED	shared chart vocabulary => same field-/tensor quotient
RV-08	Higher flavor is already closed by local $U(3)^5$ analysis	INTERFACE TO SYM9	local current data => defect/global-form closure

ID	Rival claim	Disposition	Failed implication
RV-09	Three generations follow from using 3x3 matrices	REJECTED	target dimension => cardinality proof
RV-10	The assignment certificate is circular because roles and residues share labels	REJECTED BY AUDIT	semantic compatibility => common construction

17.3 Load-bearing attacks and exact closure

Attack	Exact closure
“ $U(3)^5$ is quoted without a faithful quotient or anomaly qualification.”	T03 constructs the local commutant; T04 quotients gauged hypercharge, records the global-form-dependent center, and publishes a current-by-current anomaly table.
“Basis covariance is being called physical symmetry.”	T05 identifies basis arrows with quotient-trivial groupoid morphisms; T06 identifies fixed-point symmetry with $\text{Stab}(y)$.
“Parameter counting assumes a standard chart.”	T08 and T09 derive counts from tensor, group, and stabilizer dimensions before any CKM/PMNS parameterization.
“One Jarlskog witness is treated as universal.”	T13 indexes every CP package by target and stratum; Majorana and singular successors have separate rows.
“Majorana phases were copied from a Dirac PMNS chart.”	T17 separates field/operator targets; T09 and T18 recompute congruence quotients and singular strata.
“MFV is optional BSM structure.”	Agreed: T15 and T16 are explicitly conditional neighboring EFT theorems, never core-target facts.
“Higher flavor defeats local closure.”	T19 proves an exact interface and excludes the defect-completed target from T25; no local theorem claims the missing global data.
“AASC labels decide the physics.”	The QFT action, quotient, invariant, anomaly, and source ledgers precede the role map; role labels occur in no QFT premise.
“The occupancy certificate is circular.”	T21 publishes separate constructors, hashes, namespaces, and relation graphs with no expected map; T22 evaluates all pairs and is independently replayed.
“Three generations were assumed by the count.”	Three generations are imported as target data from an independent carrier/cardinality endpoint; T08 also states the arbitrary- n formula.
“The SYM-3 hash mismatch invalidates provenance.”	The full and manuscript-only archives are packaging-equivalent versions of one theorem authority; the user supplied the packaging reason and no theorem content changed.

Attack	Exact closure
“The certificate is just label matching.”	Opaque labels are permuted in MUT-12 without changing the result. Compatibility depends on target, kind, semantic interface, evidence, and relation fields, not the original IDs.

17.4 Adversarial-map gate

Before issuing a verdict, a referee must test each proposed objection against the exact closure above and trace the theorem chain backward. A surviving report must identify the failed premise, inference, target record, source theorem, data row, or executable check. When the map already answers the objection, the referee must explain why the cited closure fails.

17.5 Counterexample protocol

A genuine counterexample must provide:

1. an exact target fingerprint and proof that it is the same target;
2. the acting object, acted-upon object, and transformation law;
3. the physical quotient and changed observable/invariant content;
4. stabilizer, anomaly, background, scale, and stratum fields;
5. the claimed new role and proof that it is not an existing role at another locus;
6. proof that the difference is neither representative skin nor a target update;
7. the theorem and certificate row contradicted;
8. a primary-source or local-derivation certificate.

Permitted dispositions are ABSORBED EQUIVALENT, SKIN ONLY, TARGET UPDATE, INTERFACE, OPEN SUCCESSOR BURDEN, and GENUINE COUNTEREXAMPLE. The current genuine-counterexample ledger is empty.

17.6 Mutation review

The twelve mutations exercise precisely the high-risk fields: target, coverage, kind, relation, identifier uniqueness, anomaly, stratum, invariant status, evidence, higher-flavor premises, neutrino target update, and opaque-label invariance. All tests return the expected result. The full report is reproduced in Appendix J.

Exact nonclaims, downstream exports, and release conclusion

18.1 Exact nonclaims

Even successful **SYM5-T25** does not:

- determine numerical Yukawa eigenvalues, masses, mixing angles, CP phases, Wilson coefficients, or neutrino ordering;
- derive three generations from matrix dimension or parameter counting;
- make the Standard Model unique under arbitrary ultraviolet assumptions;
- state that every classical $U(1)$ factor of the $U(3)^5$ cover is an exact quantum global symmetry;
- prove that MFV is realized by nature or uniquely selected;
- provide a complete invariant ring for every seesaw, BSM flavor model, or arbitrary tensor extension;
- classify chiral, custodial, isospin, baryon/lepton, vacuum, global-form, defect, or RG symmetries outside the declared interfaces;
- close the higher-flavor two-group without the SYM-9 global-form and extended-operator target;
- promote representative fields, bases, spurions, phase charts, or matrix entries to primitive occupants;
- identify an SVD/Takagi factorization isotropy with the stabilizer of the complete tensor pair without proof;
- re-prove the Carrier SM-5 primitive-support endpoint or replace its generation-cardinality theorem;
- replace dynamics or experiment with symmetry classification;
- establish the whole Standard Model symmetry exhaustion owned by SYM-11.

18.2 Downstream exports

Module	Frozen SYM-5 export
SYM-6	Flavor-limit, basis, tensor, invariant, and anomaly records needed to distinguish electroweak flavor from QCD chiral, custodial, isospin, and infrared limits.
SYM-7	Neutrino-target, lepton-number, family-difference, anomaly, and operator-update fields for accidental-symmetry analysis.
SYM-8	Fixed-tensor stabilizers and the basis/physical mixing distinction relevant to electroweak vacuum and residual charge reconstruction.
SYM-9	Faithful-global-form placeholder, magnetic one-form background, line/defect requirement, triangle data, higher-group extension field, and monopole burdens.
SYM-10	Scale/evidence fields and invariant transport requirements for RG analysis.
SYM-11	Frozen flavor atlas, role normal form, target successors, rival ledger, finite certificate, and exact closure boundary.

18.3 Release conclusion

The completed SYM-5 object is stronger than the ARC's minimum release gate. It states exactly what the $U(3)^5$ limit means; separates the covariance group from fixed-point symmetry; constructs the quark, Dirac, and Majorana quotients; derives generic and singular parameter counts; freezes target-qualified invariant and CP packages; closes two MFV targets conditionally; preserves the higher-flavor global boundary; and produces one executable role-assignment orbit.

Final endpoint

Within the declared sharp-local target family, every admitted Standard Model flavor structure has a fixed target, transformation law, quotient status, stabilizer, invariant/CP certificate, anomaly or EFT field, AASC role, source identity, and successor rule. No second same-scope governance-inequivalent classification survives the frozen candidate and rival protocols.

The next arc action is not to reopen SYM-5's local flavor closure. It is to import these frozen records into SYM-6 and, independently, to discharge the global-form and defect fields of the higher-flavor interface in SYM-9.

Canonical target registry and successor rules

A.1 Registered target fibers

The machine-readable registry in `data/target_registry.csv` and JSON form is authoritative. The table below is generated from that registry during every build.

ID	Target	Closure status / boundary
D5-0Y-MIN	Zero-Yukawa minimal renormalizable Standard Model	CLOSED WITH GLOBAL FORM BOUNDARY: nonzero Yukawas or added neutrino operator create successor target
D5-Q-GEN	Generic three-generation quark Yukawa target	CLOSED: rank loss, degeneracy or alignment moves to D5-Q-STRAT
D5-Q-STRAT	Registered singular and degenerate quark strata	CLOSED FOR REGISTERED STRATA: unregistered orbit type requires a versioned successor row
D5-L-0NU	Minimal renormalizable lepton target with massless neutrinos	CLOSED: ν_R or Weinberg operator changes target
D5-L-DIRAC	Dirac-neutrino neighboring target	CLOSED: Majorana mass or Weinberg operator changes target
D5-L-MAJ5	Dimension-five Majorana/Weinberg effective target	CLOSED FOR REGISTERED STRATA: explicit seesaw completion is a new target
D5-MFV-Q	Quark Minimal Flavor Violation EFT target	CONDITIONAL CLOSURE: new flavor-breaking tensor creates a non-MFV successor
D5-MFV-L	Minimal-field-content lepton MFV EFT target	CONDITIONAL CLOSURE: extended-field-content seesaw MFV is a distinct target
D5-HF-IFACE	Higher-flavor/global-form interface	INTERFACE ONLY TO SYM9: SYM-9 fixes the complete defect/global-form target

A.2 Fingerprint completeness

A target identifier is not a paper label. It is the complete tuple

$$\mathcal{D} = (\text{Fields}, \text{Operators}, \text{Tensors}, \text{Basis}, \text{Kernel}, \text{Stratum}, \text{Quantum}, \text{Global}, \text{Scale}, \text{Evidence}).$$

The following fields are mandatory before a theorem may quantify over a fiber.

Field	Release meaning
field_content	Gauge-representation species, multiplicities, Higgs content, right-handed-neutrino content, and dynamical versus back-ground status.
operator_content	Renormalizable, dimension-five, or fixed EFT truncation; local versus extended operator category; and matching scale.
tensor_space	Tensor type, complex or symmetric law, rank/multiplicity restrictions, and background/dynamical status.
basis_group	Full lawful weak-basis group, its action formula, continuous kernel, and any global-form-dependent central quotient.
physical_kernel	Subgroup acting trivially on the complete tensor point or removed because it is gauged; never inferred from presentation notation alone.
neutrino_status	No mass tensor, Dirac tensor, low-energy Majorana/Weinberg tensor, or another explicit target.
global_form_status	Sharp-local closure, declared faithful quotient, line/defect category, monopole assumptions, and one-form backgrounds.
operator_locus	Fields, currents, background tensors, Wilson coefficients, lines, defects, or theory-space data on which the classification is made.
closure_status	Closed, closed for registered strata, conditional EFT closure, or interface-only.
successor_rule	The exact change that creates a versioned target rather than a same-scope rival.

A.3 Conservative transport rule

Let $u : \mathcal{D} \rightarrow \mathcal{D}'$ be a proposed transport. A frozen theorem field is inherited only when u preserves its complete dependency signature. A change to field content, operator order, tensor variance, rank/multiplicity stratum, anomaly data, invariant-completeness domain, faithful global form, line/defect category, scale, or evidence identity creates a successor record. In particular,

$$\text{massless-neutrino minimal SM} \neq \text{Dirac-neutrino SM} \neq \text{Weinberg/Majorana EFT},$$

and the shared use of a PMNS chart is not evidence of target identity.

A.4 Closure quantifier

The final sharp-local endpoint ranges over the first eight fibers with the closure qualifications printed in the registry. D5-HF-IFACE is included in the atlas and assignment certificate as an exact

interface object, but it is excluded from the claim of defect-completed closure until SYM-9 fixes the missing global data. This is a theorem boundary, not an unrecorded exception.

Import and authority registry

B.1 Upstream project imports

The import manifest contains twenty frozen project authorities. The table below records the load-bearing use of each family; the complete machine-readable rows are in `data/import_manifest.*`.

ID	Authority	Imported content	Disposition
IMP-ARC	Standard Model Symmetry ARC	SYM-5 ownership, two-track order, release gate, and original 5.1–5.7 ladder	Governing plan
IMP-SYM0	The Symmetry Status	target update, quotient trichotomy, role covariance, and finite normal-form pattern	Frozen theorem authority
IMP-SYM1	Gauge Architecture and Gauge Skin	gauge representation scaffold, faithful local product, and gauge-overlap discipline	Frozen theorem authority
IMP-SYM2	Spacetime Symmetry as Domain Anchor	active/passive and CPT premise interface	Frozen theorem authority
IMP-SYM3	Anomalies as Structural Obstructions	anomaly taxonomy and conservative obstruction transport	Full and manuscript-only packaging are one theorem authority
IMP-SYM4	Discrete Symmetry Ledger and Audit Companion	basis/rephasing quotient, generalized CP, degeneracy, and neutrino handoff	Frozen published interface
IMP-CARRIER5	Carrier SM-5 flavor transition endpoint	primitive flavor-transition and CP/holonomy support; no-promotion ledger	Support import, not a substitute for QFT
IMP-GEN	Generation-cardinality endpoint	independent three-generation governance envelope	Target datum source

ID	Authority	Imported content	Disposition
IMP-KERNEL	Kernel of Admissibility	necessity of determinate target, standing, reference, and fixation	Dependency import
IMP-FIXED	Fixed-Domain Exhaustion	fixed-scope comparison and successor discipline	Dependency import
IMP-ATS	Anchor, Tensor, and Skin	locus-sensitive role decomposition	Dependency import
IMP-CLOSURE	Same-Scope Closure	rival disposition and no-second-classifier burden	Dependency import
IMP-ROLE	Role-Occupancy Closure	independent role/residue construction and orbit theorem architecture	Certificate method
IMP-CNF5/6/7	Texture, spectrum, mixing, and phase sequence	quotient-normalized textures, rank/degeneracy, rephasing, and mixing controls	Supporting corpus machinery
IMP-MGN4/6	Yukawa compatibility and mixing standing	Yukawa slots, survival, breaking ancestry, and rephasing skin	Supporting corpus machinery
IMP-NU	Neutrino reconstruction project	physical neutrino quotient and reconstruction interface	Support only; not a full flavor-group theorem
IMP-STD	Standalone Standard Model	gauge, representation, Yukawa, charge, and anomaly spine	Supporting construction source

B.2 Authority and packaging rules

1. Substantive lineage and explicit supersession outrank filesystem copy time.
2. The post-matrix Standard Model Carrier ARC governs carrier-level flavor claims.
3. The SYM-3 full archive and Overleaf-compatible manuscript-only archive are packaging profiles of the same frozen theorem authority. Their distinct byte hashes do not create a mathematical successor.
4. SYM-4 is imported through its published manuscript and Technical Audit Companion. Every imported result is named by theorem/interface identity, not inferred from prose similarity.
5. Primary QFT literature governs framework-neutral results not derived locally.
6. Repetition across AASC manuscripts is not an independent proof certificate.
7. A local theorem may strengthen an imported result only after its target and hypotheses are reproduced exactly in the theorem registry.

B.3 Dependency direction

The dependency order is

$$K \implies A^+ \implies \text{determinate QFT target and record use,}$$

but the positive content of the commutant, orbit quotient, parameter count, invariant ring, CP criterion, and higher-group interface remains a QFT or mathematical obligation. No AASC role label occurs as a premise in the framework-neutral classification.

Primary QFT source map

C.1 Load-bearing primary sources

The following table is generated from `data/primary_source_map.*`. Each row records not only what is imported, but also the guard that prevents scope inflation.

ID	Source	Load-bearing use	Guard
QFT-01	Dambrosio2002	$U(3)^5$ presentation cover, quark Yukawa spurions and MFV hypothesis	MFV is conditional and EFT-order indexed
QFT-02	Jarlskog1985	Three-generation quark CP-odd commutator invariant and mass-difference factors	Not a universal Majorana or arbitrary-n invariant package
QFT-03	JenkinsManohar2007	Rephasing invariants for quark and lepton mixing matrices	Target-specific rephasing groups
QFT-04	JenkinsManohar2009	Invariant-ring structure, CP-odd secondary and syzygies	Specific field content and generation count
QFT-05	HananyEtAl2011	Hilbert-series count and finite flavor invariant algebra	Ring result, not numerical parameter prediction
QFT-06	CiriglianoEtAl2005	Minimal-field-content lepton MFV and separate lepton-number/flavor scales	Extended-field-content seesaw MFV is a different target
QFT-07	delAguilaZralek1996	CP phase reduction with vanishing or degenerate fermion masses	Model and target hypotheses retained
QFT-08	YuZhou2019	Majorana weak-basis CP conditions and independence subtleties	Nondegenerate low-energy Majorana target
QFT-09	YuZhou2021	Partial/full Majorana degeneracy and stratum-sensitive CP conditions	Low-energy EFT and stated degeneracy patterns

ID	Source	Load-bearing use	Guard
QFT-10	CordovaKoren2023	Higher-group mixing of flavor backgrounds with magnetic hypercharge one-form symmetry	Requires global-form / background / defect data; interface only
QFT-11	Weinberg1979	Dimension-five lepton-number-violating operator target	Low-energy operator, not a unique UV completion
QFT-12	Witten1982	Global $SU(2)$ anomaly distinction imported through SYM-3	Foundation anomaly interface

C.2 Exact source-use rules

1. D’Ambrosio–Giudice–Isidori–Strumia supplies the conventional $U(3)^5$ presentation, Yukawa transformation laws, and the quark MFV hypothesis; it does not prove that MFV is selected by the observed theory [3].
2. Jarlskog’s three-generation commutator witness is used only in its quark, nondegenerate scope and with its mass-difference prefactors visible [4].
3. Rephasing and invariant-ring sources are imported with their field content and generation number fixed; a finite ring result is not a numerical prediction [5–7].
4. The minimal-field-content lepton-MFV construction is not identified with an extended-field-content seesaw target [8].
5. Majorana CP criteria are indexed by nondegenerate, massless, partially degenerate, and fully degenerate strata; no generic list is transported across these boundaries [9–11].
6. The Weinberg operator defines a low-energy dimension-five target, not a unique ultraviolet completion [13].
7. The higher-flavor result is imported only when the faithful global form, magnetic one-form background, line/defect category, monopole assumptions, and triangle data are present [12].
8. The Witten $SU(2)$ anomaly remains an anomaly-classification import through SYM-3 and is not conflated with perturbative local anomalies [19].

C.3 Bibliographic integrity

The release bibliography records the corrected DOI for Jenkins–Manohar’s rephasing-invariant paper and the correct Acta Physica Polonica B publication data for del Aguila–Zralek. The build fails on undefined citation keys. Source identifiers are duplicated in CSV/JSON so bibliography prose is not the sole provenance surface.

Theorem registry, proof obligations, and release gates

D.1 Completed theorem registry

The 25-result registry is the canonical theorem spine. Status values distinguish local proofs, imported-and-specialized results, conditional closures, interfaces, and finite certificates.

ID	Result	Status
SYM5-T01	Target-family and fingerprint completeness	PROVED
SYM5-T02	Flavor action groupoid construction	PROVED
SYM5-T03	Zero-Yukawa commutant theorem	IMPORTED AND SPECIALIZED
SYM5-T04	Faithful quotient, Abelian and anomaly qualification	PROVED WITH GLOBAL FORM BOUNDARY
SYM5-T05	Basis-quotient descent theorem	PROVED
SYM5-T06	Fixed-tensor stabilizer theorem	PROVED
SYM5-T07	Orbit-type and degeneracy stratification	PROVED FOR REGISTERED STRATA
SYM5-T08	Quark and Dirac parameter-count theorem	PROVED
SYM5-T09	Majorana EFT parameter-count theorem	PROVED
SYM5-T10	Finite invariant algebra and orbit-separation theorem	IMPORTED AND SPECIALIZED
SYM5-T11	Representative/invariant non-collapse theorem	PROVED
SYM5-T12	Generalized-CP existence theorem	IMPORTED AND SPECIALIZED
SYM5-T13	Target- and stratum-complete CP certificate	CERTIFIED BY TARGET STRATUM REGISTRY
SYM5-T14	Spurion covariance and no-promotion theorem	PROVED
SYM5-T15	Quark MFV conditional closure	CONDITIONAL CLOSURE PROVED

ID	Result	Status
SYM5-T16	Minimal lepton MFV conditional closure	CONDITIONAL CLOSURE PROVED
SYM5-T17	Neutrino target bifurcation theorem	PROVED
SYM5-T18	Massless and degenerate neutrino stabilizer theorem	PROVED FOR REGISTERED STRATA
SYM5-T19	Higher-flavor conditional interface	CONDITIONAL INTERFACE PROVED
SYM5-T20	Complete locus-sensitive flavor role classification	PROVED
SYM5-T21	Independent role-position and residue construction	PROVED
SYM5-T22	Fiberwise unique assignment-orbit theorem	FINITE CERTIFICATE PASSED
SYM5-T23	Integrated flavor normal-form theorem	PROVED
SYM5-T24	Conservative successor and downstream transport	PROVED
SYM5-T25	Integrated flavor endpoint closure	PROVED AT DECLARED SHARP LOCAL SCOPE

D.2 Proof-obligation discharge

ID	Burden	Status	Certificate
PO-01	Target discipline	PASS	target and transport registries; T01, T17, T24
PO-02	Zero-Yukawa flavor group	PASS with global-form boundary	commutant, faithful quotient, Abelian/anomaly ledgers; T03–T04
PO-03	Basis quotient and stabilizers	PASS	basis action and stratum registries; T02, T05–T07
PO-04	Parameter counts	PASS	generic formulas and twelve stratum rows; T08–T09, T18
PO-05	Invariants and CP	PASS for declared targets/strata	invariant and CP registries; T10–T13
PO-06	Spurions and MFV	CONDITIONAL PASS	complete-spurion and EFT-order hypotheses explicit; T14–T16
PO-07	Higher flavor	INTERFACE PASS	all global-form/one-form/defect fields registered; T19

ID	Burden	Status	Certificate
PO-08	Role classification	PASS	82-row atlas and independent constructors; T20–T21
PO-09	Finite certificate and endpoint	PASS	all-pairs matrix, graph replay, one lawful orbit, mutations; T22–T25

D.3 Release-gate result

Audit-grade release gate: PASS

Every theorem and example resolves to a target/stratum ID; all framework-neutral results have local or imported certificates; the role constructors are independent of the residue constructors; all 778 same-target pairs are evaluated; the relation graphs have exactly one color-preserving directed isomorphism in every target block; all twelve mutations behave as specified; and the PDF/source package is reproducibly built and preflighted. Conditional and interface statuses are part of the theorem statements and therefore do not count as hidden open obligations.

D.4 What would force a successor

A successor release is required by any new same-scope QFT residue, a failed invariant-completeness condition, an unregistered orbit stratum, a nontrivial second assignment orbit, a genuine counterexample in the rival ledger, a change in the higher-flavor global data, or a substantive upstream theorem revision. Presentation-only rebuilding and the SYM-3 packaging split do not trigger a successor.

Canonical flavor atlas

E.1 Atlas semantics

The atlas is no longer a seed. It contains 82 canonical rows, one for every assigned QFT residue. Each row records a target, QFT object, quotient locus, quantum/source status, AASC role, role proof, rival status, nonclaim, owner, and version. The CSV and JSON forms are authoritative; this appendix is the generated human-readable projection.

Record	Target	QFT object	Role
SYM5-CHARGED-TENSOR-v1	D5-L-0NU	complex charged-lepton Yukawa matrix Y_e	tensor
SYM5-CHARGED-SPECTRUM-v1	D5-L-0NU	ordered charged-lepton singular-value triple	witness
SYM5-BASIS-QUOTIENT-v1	D5-L-0NU	quotient of one complex charged-lepton matrix	skin
SYM5-FAMILY-STABILIZER-v1	D5-L-0NU	generic diagonal family $U(1)^3$	action
SYM5-NO-PMNS-v1	D5-L-0NU	absence of neutrino mass tensor and PMNS observables	target guard
SYM5-COMMUTANT-v1	D5-0Y-MIN	$U(3)^5$ unitary multiplicity-space commutant	action
SYM5-NONABELIAN-FACTOR-v1	D5-0Y-MIN	non-Abelian $SU(3)^5$ presentation factor	action
SYM5-HYPERCHARGE-OVERLAP-v1	D5-0Y-MIN	gauged hypercharge one-parameter subgroup	boundary
SYM5-BARYON-CURRENT-v1	D5-0Y-MIN	baryon-number current with $SU(2)$ anomaly	action
SYM5-LEPTON-FAMILY-CURRENTS-v1	D5-0Y-MIN	family lepton-difference currents	action
SYM5-ABELIAN-ANOMALY-v1	D5-0Y-MIN	Abelian current anomaly and nonperturbative table	obstruction
SYM5-ZERO-YUKAWA-SUCCESSOR-v1	D5-0Y-MIN	map from the zero tensor point to nonzero-Yukawa fibers	target guard

Record	Target	QFT object	Role
SYM5-QUARK-SPURION-PAIR-v1	D5-MFV-Q	ordered pair of background spurions Y_u, Y_d	spurion
SYM5-SU3Q3-COVARIANCE-v1	D5-MFV-Q	formal $SU(3)_Q \times SU(3)_u \times SU(3)_d$ covariance	route
SYM5-OPERATOR-GRAMMAR-v1	D5-MFV-Q	complete-spurion operator construction rule	route
SYM5-DELTA F2-v1	D5-MFV-Q	Delta F=2 spurion-polynomial coefficient class	witness
SYM5-DELTA F1-v1	D5-MFV-Q	Delta F=1 spurion-polynomial coefficient class	witness
SYM5-CP-HYPOTHESIS-v1	D5-MFV-Q	declared flavor-blind CP coefficient hypothesis	boundary
SYM5-TRUNCATION-SCALE-v1	D5-MFV-Q	fixed operator dimension and matching scale	boundary
SYM5-NO-PROMOTION-v1	D5-MFV-Q	nondynamical-background certificate	no promotion
SYM5-SUCCESSOR-NONMFV-v1	D5-MFV-Q	new independent flavor-breaking tensor	target guard
SYM5-LEPTON-SPURION-PACKAGE-v1	D5-MFV-L	Y_e plus κ/g_ν spurion package	spurion
SYM5-LEPTON-FLAVOR-GROUP-v1	D5-MFV-L	minimal-field-content $SU(3)_L \times SU(3)_e$ covariance	route
SYM5-LN-SCALE-v1	D5-MFV-L	independent total-lepton-number scale Λ_{LN}	boundary
SYM5-NEUTRINO-SPURION-v1	D5-MFV-L	symmetric κ/g_ν lepton spurion	spurion
SYM5-CLFV-GRAMMAR-v1	D5-MFV-L	charged-lepton-flavor-violating operator grammar	route
SYM5-LEPTON-CP-v1	D5-MFV-L	declared leptonic CP coefficient status	boundary
SYM5-LEPTON-TRUNCATION-v1	D5-MFV-L	fixed lepton EFT truncation and matching scale	boundary
SYM5-LEPTON-NO-PROMOTION-v1	D5-MFV-L	nondynamical lepton-spurion certificate	no promotion
SYM5-SEESAW-BOUNDARY-v1	D5-MFV-L	new heavy-neutrino/seesaw field content	target guard
SYM5-QUARK-PAIR-v1	D5-Q-GEN	pair (Y_u, Y_d) of complex 3x3 tensors	tensor
SYM5-QUARK-BASIS-ACTION-v1	D5-Q-GEN	$U(3)_Q \times U(3)_u \times U(3)_d$ field-basis action	skin
SYM5-BARYON-KERNEL-v1	D5-Q-GEN	common $U(1)_B$ kernel of the generic pair action	action
SYM5-QUARK-ORBIT-v1	D5-Q-GEN	ten-dimensional orbit of a generic quark Yukawa pair	quotient

Record	Target	QFT object	Role
SYM5-QUARK-SPECTRA-v1	D5-Q-GEN	six singular values of Y_u and Y_d	witness
SYM5-CKM-MODULI-v1	D5-Q-GEN	three independent CKM mixing moduli	witness
SYM5-CKM-REPHASING-v1	D5-Q-GEN	generic CKM rephasing-equivalence class	quotient
SYM5-QUARK-CP-v1	D5-Q-GEN	CP-odd commutator/Jarlskog invariant	witness
SYM5-QUARK-INVARIANT-RING-v1	D5-Q-GEN	finite primary/secondary quark invariant algebra	witness
SYM5-QUARK-CHART-v1	D5-Q-GEN	chosen CKM/Yukawa matrix coordinates	skin
SYM5-DIRAC-PAIR-v1	D5-L-DIRAC	pair (Y_e, Y_{ν}) of complex 3x3 tensors	tensor
SYM5-DIRAC-BASIS-ACTION-v1	D5-L-DIRAC	$U(3)_L \times U(3)_e \times U(3)_{\nu}$ field-basis action	skin
SYM5-LEPTON-KERNEL-v1	D5-L-DIRAC	common total-lepton $U(1)$ fixed action	action
SYM5-DIRAC-ORBIT-v1	D5-L-DIRAC	ten-dimensional generic Dirac lepton orbit	quotient
SYM5-DIRAC-SPECTRA-v1	D5-L-DIRAC	six charged-lepton and neutrino singular values	witness
SYM5-PMNS-MODULI-v1	D5-L-DIRAC	three PMNS mixing moduli	witness
SYM5-DIRAC-REPHASING-v1	D5-L-DIRAC	generic PMNS rephasing-equivalence class	quotient
SYM5-DIRAC-CP-v1	D5-L-DIRAC	Dirac CP-odd rephasing invariant	witness
SYM5-TOTAL-LEPTON-NUMBER-v1	D5-L-DIRAC	physical total-lepton-number current in the Dirac target	action
SYM5-DIRAC-CHART-v1	D5-L-DIRAC	chosen Dirac PMNS parametrization	skin
SYM5-FAITHFUL-GLOBAL-FORM-v1	D5-HF-IFACE	fixed faithful Standard Model global gauge form	target guard
SYM5-MAGNETIC-BACKGROUND-v1	D5-HF-IFACE	magnetic hypercharge one-form two-form background	interface
SYM5-LINE-CATEGORY-v1	D5-HF-IFACE	genuine line and defect operator category	target guard
SYM5-FLAVOR-BACKGROUNDS-v1	D5-HF-IFACE	zero-form flavor background gauge fields	interface
SYM5-TRIANGLE-COEFFICIENTS-v1	D5-HF-IFACE	mixed flavor-hypercharge triangle coefficients	witness
SYM5-TWO-GROUP-EXTENSION-v1	D5-HF-IFACE	higher-group extension/Postnikov datum	interface
SYM5-MONOPOLE-ASSUMPTION-v1	D5-HF-IFACE	dynamical-monopole presence/absence flag	boundary

Record	Target	QFT object	Role
SYM5-HIGHER-SPURION-v1	D5-HF-IFACE	Yukawa higher-group spurion transformation	spurion
SYM5-SYM9-HANDOFF-v1	D5-HF-IFACE	versioned downstream SYM-9 interface packet	interface
SYM5-LOCAL-EXCLUSION-v1	D5-HF-IFACE	flag excluding defect target from local closure quantifier	target guard
SYM5-STRATUM-FINGERPRINT-v1	D5-Q-STRAT	registered orbit-type fingerprint	target guard
SYM5-RANK2-UP-v1	D5-Q-STRAT	rank-two up-type Yukawa tensor pair	tensor
SYM5-U21-TENSOR-v1	D5-Q-STRAT	2+1 degenerate up-type Yukawa tensor pair	tensor
SYM5-COMMUTING-PAIR-v1	D5-Q-STRAT	commuting Hermitian pair H_u, H_d	tensor
SYM5-STRATUM-STABILIZER-v1	D5-Q-STRAT	full-pair stabilizer conjugacy-class table	action
SYM5-STRATUM-DIMENSION-v1	D5-Q-STRAT	orbit-space dimension calculation	witness
SYM5-PHASE-SURVIVAL-v1	D5-Q-STRAT	physical phase-count table	witness
SYM5-STRATUM-CP-v1	D5-Q-STRAT	stratum-indexed quark CP test	witness
SYM5-STRATUM-SUCCESSOR-v1	D5-Q-STRAT	generic-to-singular target transport	target guard
SYM5-SINGULAR-CHART-v1	D5-Q-STRAT	chosen singular-value/CKM chart	skin
SYM5-STRATUM-COMPLETENESS-v1	D5-Q-STRAT	registered-strata-only closure flag	boundary
SYM5-MAJORANA-CHARGED-TENSOR-v1	D5-L-MAJ5	complex charged-lepton Yukawa matrix	tensor
SYM5-KAPPA-TENSOR-v1	D5-L-MAJ5	complex symmetric Weinberg coefficient kappa	tensor
SYM5-MAJORANA-BASIS-ACTION-v1	D5-L-MAJ5	$U(3)_L \times U(3)_e$ basis/congruence action	skin
SYM5-MAJORANA-ORBIT-v1	D5-L-MAJ5	twelve-dimensional generic Majorana orbit	quotient
SYM5-MAJORANA-SPECTRA-v1	D5-L-MAJ5	charged-lepton and Majorana mass spectra	witness
SYM5-MAJORANA-MODULI-v1	D5-L-MAJ5	PMNS quotient mixing moduli	witness
SYM5-MAJORANA-PHASES-v1	D5-L-MAJ5	Dirac-plus-Majorana rephasing class	quotient
SYM5-MAJORANA-CP-v1	D5-L-MAJ5	stratum-indexed weak-basis CP invariant package	witness

Record	Target	QFT object	Role
SYM5-MAJORANA-STRATA-v1	D5-L-MAJ5	rank/degeneracy Takagi stratum registry	target guard
SYM5-LN-BREAKING-v1	D5-L-MAJ5	Delta L=2 Weinberg-operator status	obstruction
SYM5-MAJORANA-CHART-v1	D5-L-MAJ5	chosen Majorana PMNS/-Takagi chart	skin

E.2 Role vocabulary

The atlas uses the following controlled loci.

Role	Operational criterion
action	Nontrivial action or current structure on the declared physical quotient or limit target.
skin	A lawful weak-basis or chart change preserving all quotient-stable content.
tensor	A standing-bearing Yukawa, mass, or response tensor at the declared locus.
spurion	A nondynamical background tensor with a declared covariance law and breaking ancestry.
witness	A spectrum, invariant, CP diagnostic, quotient coordinate, or operator consequence.
route	A conditional EFT construction grammar or effective covariance route.
obstruction	Anomaly, incompatibility, or matching datum blocking an otherwise proposed status.
boundary	Scale, global-form, coefficient, or premise field that limits the theorem quantifier.
interface	Exact export whose completion belongs to another target/module.
target_guard	A target-exclusion or successor trigger preventing hidden field/operator changes.
no_promotion	Certificate that representation or bookkeeping data do not acquire primitive/dynamical status.

E.3 Atlas covariance

Admitted representative changes may alter the displayed matrix, labels, phase chart, or diagonalization route but not the target ID, quotient invariant package, anomaly status, role locus, or evidence identity. A change in any of those frozen fields either fails covariance or creates a successor row.

Formalization boundary and executable semantics

F.1 What is executable

The release mechanizes the finite audit surface rather than the interacting Standard Model. The executable objects are:

- typed target, theorem, stratum, invariant, anomaly, spurion, transport, rival, and nonclaim registries;
- separately authored role-burden and QFT-residue constructors;
- a complete same-target eligibility matrix;
- directed relation graphs for both constructors;
- the unique canonical assignment and its automorphism quotient;
- twelve semantic mutation tests;
- independent NetworkX replay;
- schema, cross-reference, source, scope, dimension, and count validation.

F.2 Constructor datatypes

A role burden is represented by

$$(r, \mathcal{D}, k, \sigma, e, d),$$

where r is an opaque identifier, k a required role kind, σ a semantic interface signature, e an evidence requirement, and d an independently written description. A QFT residue is represented by

$$(q, \mathcal{D}, k', \sigma', S, Q, d'),$$

where S is a nonempty source/derivation field and Q is the quotient-status field. Neither constructor stores an expected cross-map.

Eligibility is the conjunction

$$\text{Elig}(r, q) \iff \mathcal{D}_r = \mathcal{D}_q \wedge k_r = k_q \wedge \sigma_r = \sigma_q \wedge S_q \neq \emptyset.$$

Relational compatibility additionally requires every directed role edge to map to a directed QFT edge. The certificate is therefore stronger than a count comparison but weaker than an unsupervised discovery theorem: the semantic interfaces are intentionally declared audit types.

F.3 Constructor independence

The two canonical JSON files have disjoint namespaces, distinct paths, distinct hashes, different descriptions, and no expected-map field. Their relation orders are separately authored. Opaque-label mutation MUT-12 renames every role and residue identifier while preserving semantic and relational content; the unique result survives.

F.4 Formal proof boundary

A future Lean audit can encode target fingerprints, well-formedness, finite eligibility, graph preservation, automorphism action, and the singleton-orbit theorem. It need not claim a machine-checked derivation of path integrals, anomaly polynomials, invariant-ring theorems, or the nonperturbative Standard Model. Those remain named imported interfaces unless separately formalized.

F.5 Independent replay

The primary validator constructs the all-pairs matrix and checks the canonical assignment. The second implementation uses `networkx.algorithms.isomorphism.DiGraphMatcher` on colored directed graphs. Each target block has exactly one color-preserving isomorphism; their product therefore has one complete assignment and one lawful orbit.

Orbit-type and singular-stratum registry

G.1 Dimension convention

For a compact group action on a smooth orbit-type stratum,

$$\dim \mathcal{Q}_{\text{phys}}^{(\lambda)} = \dim \mathcal{Y}_\lambda - \dim G^{\text{basis}} + \dim \text{Stab}_{G^{\text{basis}}}(y_\lambda).$$

The tensor-stratum dimension already includes rank and eigenvalue-multiplicity restrictions. Factorization isotropy (such as an $O(2)$ freedom in a Takagi factorization) is listed separately from the stabilizer of the complete tensor pair.

Stratum	Target	dim T	dim G	dim Q	Physical decomposition
Q-GEN	D5-Q-GEN	36	27	10	6 masses + 3 angles + 1 CP phase
Q-U0	D5-Q-STRAT	34	27	9	5 masses + 3 angles + 1 CP phase
Q-U21	D5-Q-STRAT	33	27	7	5 distinct masses + 2 real mixing angles; no physical CKM phase
Q-COMM	D5-Q-STRAT	30	27	6	6 masses; no physical mixing
L0-GEN	D5-L-0NU	18	18	3	3 charged-lepton masses; no physical PMNS data
LD-GEN	D5-L-DIRAC	36	27	10	6 masses + 3 angles + 1 Dirac phase
LD-N0	D5-L-DIRAC	34	27	9	5 masses + 3 angles + 1 Dirac phase
LD-N21	D5-L-DIRAC	33	27	7	5 distinct masses + 2 real mixing angles; no Dirac phase
LM-GEN	D5-L-MAJ5	30	18	12	6 masses + 3 angles + 3 CP phases
LM-M0	D5-L-MAJ5	28	18	10	5 masses + 3 angles + 2 CP phases

Stratum	Target	dim T	dim G	dim Q	Physical decomposition
LM-M21	D5-L-MAJ5	28	18	10	5 distinct masses + 5 mixing coordinates, including 2 CP phases under the cited convention
LM-M3	D5-L-MAJ5	25	18	7	4 spectral parameters + 3 quotient mixing coordinates; no generic PMNS chart is privileged

G.2 Quark strata

- **Q-U0**: one vanishing up-type singular value removes one mass parameter but does not generically remove the CKM phase. The quotient therefore has nine parameters.
- **Q-U21**: a $2 + 1$ up-type degeneracy reduces the tensor stratum and removes one physical mixing coordinate and the single three-generation CP phase, leaving five distinct masses and two real angles.
- **Q-COMM**: commuting Hermitian combinations are simultaneously diagonalizable at the registered locus, so physical mixing disappears and only six masses remain.

G.3 Dirac-lepton strata

The Dirac rows reproduce the quark orbit arithmetic with total lepton number replacing baryon number. One massless neutrino retains a generic Dirac phase; a $2 + 1$ degeneracy removes the Dirac phase and one angle.

G.4 Majorana strata

The low-energy Majorana quotient is a congruence problem for (Y_e, κ) rather than a second Dirac biunitary problem. For three generations:

- generic nondegenerate masses give twelve parameters and three independent CP phases;
- one massless neutrino gives ten parameters and two CP phases;
- a partial $2 + 1$ degeneracy gives ten parameters, with three mixing angles and two CP phases in the cited convention;
- complete degeneracy gives seven parameters, with two mixing angles and one CP phase after the enhanced real orthogonal freedom is used.

These are stratum-specific quotient statements; no generic PMNS chart is granted primitive or universal status [11].

G.5 Registration boundary

The registry is complete for the twelve named strata, not for every algebraic orbit type of arbitrary tensor extensions. An unregistered rank/multiplicity/alignment pattern is a versioned successor burden, not evidence against a theorem that quantifies only over the registered strata.

Invariant and generalized-CP registries

H.1 Invariant normal forms

ID	Target / str.	Invariant object	Completeness
INV-Q-10	D5-Q-GEN / Q-GEN	$\text{tr}(\mathbf{H}_u^k), \text{tr}(\mathbf{H}_d^k),$ $k=1,2,3$	generic and all strata as spectral data
INV-Q-11	D5-Q-GEN / Q-GEN	$\text{tr}(\mathbf{H}_u \mathbf{H}_d)$	part of finite generic separating package
INV-Q-12	D5-Q-GEN / Q-GEN	$\text{tr}(\mathbf{H}_u^2 \mathbf{H}_d)$	part of finite generic separating package
INV-Q-13	D5-Q-GEN / Q-GEN	$\text{tr}(\mathbf{H}_u \mathbf{H}_d^2)$	part of finite generic separating package
INV-Q-14	D5-Q-GEN / Q-GEN	$\text{tr}(\mathbf{H}_u^2 \mathbf{H}_d^2)$	part of finite generic separating package
INV-Q-CP	D5-Q-GEN / Q-GEN, Q-U0, Q-U21, Q-COMM	$\text{Im tr}([\mathbf{H}_u, \mathbf{H}_d]^3)$ (equivalently the three-generation Jarlskog commutator determinant up to normalization)	three-generation quark CP criterion; interpretation is stratum-sensitive
INV-LD-CP	D5-L-DIRAC / LD-GEN, LD-N0, LD-N21	Dirac analogue $\text{Im tr}([\mathbf{H}_e, \mathbf{H}_\nu]^3)$	three-generation Dirac target with target hypotheses
INV-LM-WB	D5-L-MAJ5 / LM-GEN	registered weak-basis invariants I_1, I_2, I_4 (source convention)	hypothesis-indexed; not transported to degeneracy
INV-LM-21	D5-L-MAJ5 / LM-M21	partial-degeneracy invariant pair/package in source convention	necessary/sufficient under cited stratum assumptions
INV-LM-M0	D5-L-MAJ5 / LM-M0	massless-neutrino successor invariant package	stratum-specific
INV-LM-M3	D5-L-MAJ5 / LM-M3	complete-degeneracy invariant package	stratum-specific; no generic chart transport

H.2 Quark reconstruction

For the generic three-generation quark target define

$$H_u = Y_u Y_u^\dagger, \quad H_d = Y_d Y_d^\dagger.$$

The spectral traces and determinants recover the two unordered singular-value spectra. Mixed traces recover the relative eigenspace orientation, while the CP-odd commutator invariant

$$I_{\text{CP}}^q = \text{Im tr}[H_u, H_d]^3$$

is proportional to the Jarlskog invariant multiplied by all relevant up- and down-sector squared-mass differences [4]. Hence it vanishes under CP conservation, a two-generation reduction, or the relevant degeneracy. It is not used as a complete Majorana certificate.

H.3 Generalized CP

For every tensor target, CP conservation means the existence of unitary maps compatible with the complete target law. Schematically,

$$X_L^\dagger Y_e X_e = Y_e^*, \quad X_L^\dagger Y_\nu X_\nu = Y_\nu^*$$

in the Dirac target, while the Weinberg/Majorana tensor satisfies the congruence law

$$X_L^\dagger \kappa X_L^* = \kappa^*.$$

A complex coefficient in a chosen basis is neither necessary nor sufficient for physical CP violation. The certificate is an existence theorem or a target-complete invariant package.

H.4 Completeness policy

1. Finite generation of a compact invariant ring is separated from a claim that one displayed subset separates every singular orbit.
2. Generic nondegenerate quark and lepton packages are versioned independently from massless and degenerate successor packages.
3. Majorana CP conditions record both necessity and sufficiency hypotheses. The number of CP phases is not automatically the number of independent polynomial conditions needed globally [10].
4. A chart angle or phase becomes a physical quotient coordinate only after the relevant rephasing, permutation, and degeneracy stabilizers are removed.
5. Numerical values remain empirical/downstream data.

H.5 Representative/invariant firewall

The invariant registry rejects raw matrix entries, unquotiented phases, arbitrary eigenvector phases, diagonalization conventions, and parameterization orderings. MUT-08 tests this firewall by inserting a chart-dependent phase into the invariant list; the validator rejects the mutation.

Canonical role–residue assignment certificate

I.1 Global counts

The independent constructors contain 82 role burdens and 82 QFT residues. Pair evaluation is restricted only by target identity, so the number of evaluated pairs is

$$5^2 + 7^2 + 9^2 + 9^2 + 10^2 + 10^2 + 10^2 + 11^2 + 11^2 = 778.$$

Exactly 82 pairs satisfy the target, kind, semantic-signature, and evidence predicates. Every role has one eligible residue, every residue is used once, all directed relations are preserved, and the lawful automorphism group is trivial after target/kind/signature coloring. Thus there is one complete assignment and one lawful orbit.

I.2 Canonical assignment

Role ID	Residue ID	Signature
ROLE-D5-L-0NU-01	QFT-D5-L-0NU-05	charged_tensor
ROLE-D5-L-0NU-02	QFT-D5-L-0NU-02	charged_spectrum
ROLE-D5-L-0NU-03	QFT-D5-L-0NU-01	basis_quotient
ROLE-D5-L-0NU-04	QFT-D5-L-0NU-03	family_stabilizer
ROLE-D5-L-0NU-05	QFT-D5-L-0NU-04	no_pmns
ROLE-D5-0Y-MIN-01	QFT-D5-0Y-MIN-02	commutant
ROLE-D5-0Y-MIN-02	QFT-D5-0Y-MIN-03	nonabelian_factor
ROLE-D5-0Y-MIN-03	QFT-D5-0Y-MIN-01	hypercharge_overlap
ROLE-D5-0Y-MIN-04	QFT-D5-0Y-MIN-04	baryon_current
ROLE-D5-0Y-MIN-05	QFT-D5-0Y-MIN-05	lepton_family_currents
ROLE-D5-0Y-MIN-06	QFT-D5-0Y-MIN-06	abelian_anomaly
ROLE-D5-0Y-MIN-07	QFT-D5-0Y-MIN-07	zero_yukawa_successor
ROLE-D5-MFV-Q-01	QFT-D5-MFV-Q-03	quark_spurion_pair
ROLE-D5-MFV-Q-02	QFT-D5-MFV-Q-02	su3q3_covariance
ROLE-D5-MFV-Q-03	QFT-D5-MFV-Q-04	operator_grammar
ROLE-D5-MFV-Q-04	QFT-D5-MFV-Q-05	deltaf2

Role ID	Residue ID	Signature
ROLE-D5-MFV-Q-05	QFT-D5-MFV-Q-01	deltafl
ROLE-D5-MFV-Q-06	QFT-D5-MFV-Q-06	cp_hypothesis
ROLE-D5-MFV-Q-07	QFT-D5-MFV-Q-07	truncation_scale
ROLE-D5-MFV-Q-08	QFT-D5-MFV-Q-08	no_promotion
ROLE-D5-MFV-Q-09	QFT-D5-MFV-Q-09	successor_nonmfv
ROLE-D5-MFV-L-01	QFT-D5-MFV-L-03	lepton_spurion_package
ROLE-D5-MFV-L-02	QFT-D5-MFV-L-01	lepton_flavor_group
ROLE-D5-MFV-L-03	QFT-D5-MFV-L-04	ln_scale
ROLE-D5-MFV-L-04	QFT-D5-MFV-L-02	neutrino_spurion
ROLE-D5-MFV-L-05	QFT-D5-MFV-L-05	clfv_grammar
ROLE-D5-MFV-L-06	QFT-D5-MFV-L-06	lepton_cp
ROLE-D5-MFV-L-07	QFT-D5-MFV-L-07	lepton_truncation
ROLE-D5-MFV-L-08	QFT-D5-MFV-L-08	lepton_no_promotion
ROLE-D5-MFV-L-09	QFT-D5-MFV-L-09	seesaw_boundary
ROLE-D5-Q-GEN-01	QFT-D5-Q-GEN-05	quark_pair
ROLE-D5-Q-GEN-02	QFT-D5-Q-GEN-06	quark_basis_action
ROLE-D5-Q-GEN-03	QFT-D5-Q-GEN-07	baryon_kernel
ROLE-D5-Q-GEN-04	QFT-D5-Q-GEN-01	quark_orbit
ROLE-D5-Q-GEN-05	QFT-D5-Q-GEN-02	quark_spectra
ROLE-D5-Q-GEN-06	QFT-D5-Q-GEN-08	ckm_moduli
ROLE-D5-Q-GEN-07	QFT-D5-Q-GEN-03	ckm_rephasing
ROLE-D5-Q-GEN-08	QFT-D5-Q-GEN-09	quark_cp
ROLE-D5-Q-GEN-09	QFT-D5-Q-GEN-10	quark_invariant_ring
ROLE-D5-Q-GEN-10	QFT-D5-Q-GEN-04	quark_chart
ROLE-D5-L-DIRAC-01	QFT-D5-L-DIRAC-02	dirac_pair
ROLE-D5-L-DIRAC-02	QFT-D5-L-DIRAC-03	dirac_basis_action
ROLE-D5-L-DIRAC-03	QFT-D5-L-DIRAC-04	lepton_kernel
ROLE-D5-L-DIRAC-04	QFT-D5-L-DIRAC-05	dirac_orbit
ROLE-D5-L-DIRAC-05	QFT-D5-L-DIRAC-06	dirac_spectra
ROLE-D5-L-DIRAC-06	QFT-D5-L-DIRAC-07	pmns_moduli
ROLE-D5-L-DIRAC-07	QFT-D5-L-DIRAC-01	dirac_rephasing
ROLE-D5-L-DIRAC-08	QFT-D5-L-DIRAC-08	dirac_cp
ROLE-D5-L-DIRAC-09	QFT-D5-L-DIRAC-09	total_lepton_number
ROLE-D5-L-DIRAC-10	QFT-D5-L-DIRAC-10	dirac_chart

Role ID	Residue ID	Signature
ROLE-D5-HF-IFACE-01	QFT-D5-HF-IFACE-02	faithful_global_form
ROLE-D5-HF-IFACE-02	QFT-D5-HF-IFACE-01	magnetic_background
ROLE-D5-HF-IFACE-03	QFT-D5-HF-IFACE-03	line_category
ROLE-D5-HF-IFACE-04	QFT-D5-HF-IFACE-04	flavor_backgrounds
ROLE-D5-HF-IFACE-05	QFT-D5-HF-IFACE-05	triangle_coefficients
ROLE-D5-HF-IFACE-06	QFT-D5-HF-IFACE-06	two_group_extension
ROLE-D5-HF-IFACE-07	QFT-D5-HF-IFACE-07	monopole_assumption
ROLE-D5-HF-IFACE-08	QFT-D5-HF-IFACE-08	higher_spurion
ROLE-D5-HF-IFACE-09	QFT-D5-HF-IFACE-09	sym9_handoff
ROLE-D5-HF-IFACE-10	QFT-D5-HF-IFACE-10	local_exclusion
ROLE-D5-Q-STRAT-01	QFT-D5-Q-STRAT-01	stratum_fingerprint
ROLE-D5-Q-STRAT-02	QFT-D5-Q-STRAT-02	rank2_up
ROLE-D5-Q-STRAT-03	QFT-D5-Q-STRAT-03	u21_tensor
ROLE-D5-Q-STRAT-04	QFT-D5-Q-STRAT-04	commuting_pair
ROLE-D5-Q-STRAT-05	QFT-D5-Q-STRAT-05	stratum_stabilizer
ROLE-D5-Q-STRAT-06	QFT-D5-Q-STRAT-06	stratum_dimension
ROLE-D5-Q-STRAT-07	QFT-D5-Q-STRAT-07	phase_survival
ROLE-D5-Q-STRAT-08	QFT-D5-Q-STRAT-08	stratum_cp
ROLE-D5-Q-STRAT-09	QFT-D5-Q-STRAT-09	stratum_successor
ROLE-D5-Q-STRAT-10	QFT-D5-Q-STRAT-10	singular_chart
ROLE-D5-Q-STRAT-11	QFT-D5-Q-STRAT-11	stratum_completeness
ROLE-D5-L-MAJ5-01	QFT-D5-L-MAJ5-01	majorana_charged_tensor
ROLE-D5-L-MAJ5-02	QFT-D5-L-MAJ5-02	kappa_tensor
ROLE-D5-L-MAJ5-03	QFT-D5-L-MAJ5-03	majorana_basis_action
ROLE-D5-L-MAJ5-04	QFT-D5-L-MAJ5-04	majorana_orbit

Role ID	Residue ID	Signature
ROLE-D5-L-MAJ5-05	QFT-D5-L-MAJ5-05	majorana_spectra
ROLE-D5-L-MAJ5-06	QFT-D5-L-MAJ5-06	majorana_moduli
ROLE-D5-L-MAJ5-07	QFT-D5-L-MAJ5-07	majorana_phases
ROLE-D5-L-MAJ5-08	QFT-D5-L-MAJ5-08	majorana_cp
ROLE-D5-L-MAJ5-09	QFT-D5-L-MAJ5-09	majorana_strata
ROLE-D5-L-MAJ5-10	QFT-D5-L-MAJ5-10	ln_breaking
ROLE-D5-L-MAJ5-11	QFT-D5-L-MAJ5-11	majorana_chart

I.3 What the certificate proves

The certificate proves uniqueness relative to the frozen target blocks, role kinds, semantic interfaces, evidence requirements, and relation graphs. It detects missing roles, duplicated identifiers, cross-target substitutions, kind/signature mismatches, missing sources, and relation failures. It does not derive the semantic types from raw prose or replace the QFT proofs that justify each residue.

I.4 Independent replay result

The primary validator and the NetworkX replay agree target by target. The replay finds one colored directed graph isomorphism in every block and a product count of one. The JSON report records node counts, edge counts, canonical-map agreement, and engine identity.

Mutation and regression tests

J.1 Purpose

A passing certificate can still be vacuous if its validator never rejects plausible failures. The release therefore applies twelve controlled mutations to copies of the canonical data. Each mutation targets a load-bearing semantic field or a label-invariance requirement.

Test	Mutation	Expected signal	Result
MUT-01	Replace one residue target ID	eligibility/cross-target rejection	PASS
MUT-02	Delete one role burden	coverage rejection	PASS
MUT-03	Change one residue kind	kind/eligibility rejection	PASS
MUT-04	Delete one QFT relation edge	relational-compatibility rejection	PASS
MUT-05	Duplicate one role identifier	uniqueness rejection	PASS
MUT-06	Mark anomalous $B + L$ as exactly conserved quantum mechanically	anomaly contradiction	PASS
MUT-07	Apply a generic Majorana invariant row to an unregistered stratum	stratum-scope rejection	PASS
MUT-08	Add a chart-dependent phase as an invariant	representative/invariant rejection	PASS
MUT-09	Remove evidence from an assigned residue	missing-evidence rejection	PASS
MUT-10	Delete the faithful-global-form higher-flavor field	incomplete-interface rejection	PASS
MUT-11	Reclassify a neutrino field-content change as representative transport	hidden-target-change rejection	PASS
MUT-12	Permute every opaque role/residue label and relation endpoint	acceptance under label invariance	PASS

J.2 Interpretation

The first eleven tests demonstrate negative sensitivity; the twelfth demonstrates that uniqueness does not depend on the original identifier strings. The machine-readable report records the observed signals and returns nonzero if any expected result is absent.

Reproducibility, file layout, and artifact verification

K.1 One-command rebuild

A clean checkout is built with

```
./build.sh
```

The command regenerates canonical data and LaTeX projections, validates all registries, replays the graph certificate, runs mutations, compiles the bibliography and manuscript, checks the log, preflights the PDF, records metadata and fonts, and generates a SHA-256 manifest.

K.2 Controlled project layout

Path	Authority
data/	Canonical target, theorem, stratum, anomaly, invariant, CP, spurion, transport, atlas, rival, nonclaim, source, eligibility, assignment, and mutation records.
constructors/	Separately authored role-burden and QFT-residue inputs and their relation graphs.
schemas/	Machine-readable atlas-row constraints.
tools/	Data generation, release validation, independent replay, mutation tests, and manifest generation.
generated/	Build-generated LaTeX projections; never edited as primary authority.
sections/ appendices/	Human-readable theorem paper and audit companion material.
audit/	Validator, replay, mutation, source-audit, PDF-preflight, metadata, font, and visual-inspection reports.
build/	Final PDF and release manifest generated from a clean build.

K.3 Artifact checks

The release gate requires: no undefined citations or references; no overfull boxes; all fonts embedded; no encryption; valid page boxes; successful rendering of every page; no clipped text, overlapping tables, broken glyphs, or blank substantive pages; exact validator/replay/mutation passes; and a manifest covering every distributed source and generated canonical artifact.

K.4 Packaging note

The project ZIP contains the manuscript sources, bibliography, canonical data, constructors, validators, audit reports, and final PDF. It deliberately excludes the large upstream source corpus and primary-source PDFs; provenance is retained through import/source manifests and bibliographic identifiers. This keeps the release rebuildable and suitable for Overleaf while respecting source ownership and avoiding redundant archives.

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